Answers

Final answers only, for checking your own work. Where a question asks you to prove or to show a result, the answer is simply **Proof** — there is nothing to check but your own argument. Full worked solutions are in the markschemes, which are published separately so that reading an answer here cannot spoil a question you have not yet attempted.

Chapter 1 · Exponents & Logarithms

YOUR TURN

1. (a) $3c^5$ (b) $81x^8y^{-12}$ (c) $3a^3b^4$ (d) 1
2. (a) $2x^2-4x$ (b) $4a^4-3b^2$
3. $\frac{a^9}{b^5}$
4. (a) 9 (b) $\frac{1}{27}$ (c) $\frac{9}{4}$ (d) $\frac{1}{4}$
5. (a) $x^{3/4}$ (b) $x^{-2/3}$ (c) $x^{1/2}$ (d) x
6. 16
7. (a) $3\sqrt{2}$ (b) $5\sqrt{3}$
8. $3+\sqrt{5}$
9. 4.7×10^{-5}
10. $320\,000$
11. 7.2×10^{-2}
12. 3.0×10^{5}
13. 5.49×10^{4}
14. 1.83×10^{3}
15. (a) 5 (b) -2 (c) 9 (d) 5
16. (a) 4 (b) 125
17. (a) 15 (b) $x=4$
18. (a) $\log_3 35$ (b) 3 (c) $\log_a(x^3y)$ (d) $\ln\left(\frac{a^2c}{b^3}\right)$
19. (a) $2\log_5 a-\log_5 b$ (b) $2+p$ (c) $3p+2q$ (d) 2.682 (e) $\frac{\ln 9}{\ln 4}$ (f) $\frac{\log_2 x}{3}$
20. (a) $\log_a(x-1)$ (b) $\frac{1}{\log_b a}$
21. 3.56
22. (a) $\frac{3}{2}$ (b) 3
23. (a) 3.32 (b) 1.73
24. (a) $\ln 3$ (b) 2 (c) 1

END-OF-CHAPTER PROBLEMS

1. 4
2. $\log_2 k-\log_2 7$
3. (a) $\frac{2y^{10}}{x^{10}}$ (b) 3 (c) $a+2+a^{-1}$
4. (a) 81.2 (3 s.f.) (b) 6.04×10^{24} kg
5. 3
6. (a) 11 (b) 4 (c) -7
7. (a) $p+q$ (b) $p+\frac{q}{2}$ (c) $2p-2q$
8. 16
9. 3
10. (a) $2a$ (b) $a/2$ (c) a (d) $3/a$
11. $\frac{7}{4}$
12. $x=4$
13. $4+4\sqrt{3}$
14. 0.183
15. -0.68; both lie in the domain
16. 86.6
17. 1. **No solution.**
18. $\frac{\ln 245}{\ln(7/5)}$
19. $x=\ln 5$
20. 2
21. **19 h 56 min**
22. 14.7. First full year after crossing: **2035**
23. 7.16. Answer: **after 8 years**
24. 10^4. Solution A is **10 000** times more acidic
25. $\frac{1}{2}\cdot g(x)-\frac{1}{2}$
26. 33.2 years

27. (a) 7.14×10^4 times wider (b) 5.5×10^{61} (c) 1.05×10^2 m (d) 42 it is 4.40×10^8 m.) (e) Doubling each fold gives exponential, not linear, growth

28. (a) $P(1.04)^t$ (b) 1.03970 (5 d.p.) (c) Bank A gives more, by $0.00420P$, about 0.42% of P (d) 17.7 years (e) Proof

29. (a) Proof (b) Proof (c) 16 (d) 3

CHALLENGE QUESTIONS

30. $x = 2$

31. 83 dB

32. (a) $\lfloor \log_{10} N \rfloor + 1$ (b) 302 (c) $0.1072 \in [0, 0.3010)$ (d) Proof (e) 1 is very close; 4.6% of 30 is 1.4, so 0 is unsurprising (f) Counts support the law; 30 is too small a sample

Chapter 2 · Sequences & Series

YOUR TURN

1. (a) 27 (b) -44 (c) 31 (d) -68

2. (a) 4 (b) 8 (c) 3 (d) 6

3. 33

4. 2

5. 101 is the 17th term

6. (a) 222 (b) 1050 (c) 525 (d) 1200

7. 1056

8. 3366

9. $-13)$

10. 39

11. $8n - 5$

12. (a) 486 (b) $\frac{3}{2}$ (c) 4 (d) -384

13. (a) 2 (b) 3 (c) $\pm\frac{1}{3}$ (both real values are valid) (d) 5

14. 9

15. 2

16. 3

17. (a) 3280 (b) 8.9998 (c) 252 (d) -21

18. 3

19. 6

20. 3069

21. 1

22. (a) $\sum_{k=1}^{5} 5k$ (b) $\sum_{k=1}^{5} 2^{k+1}$ (or $\sum_{k=1}^{5} 4 \cdot 2^{k-1}$) (c) $\sum_{k=1}^{10} 3k$ (d) $\sum_{k=1}^{5} k^2$

23. 51

24. 242

25. 65

26. $\frac{5n(n+1)}{2}$

27. (a) 18 (b) 25 (c) $\frac{27}{2}$ (d) $\frac{32}{5}$

28. (a) 16 (b) $\frac{1}{3}$ (c) 9 (d) $-\frac{1}{4}$

29. $\frac{5}{9}$

30. 1.5, so $|r| \geq 1$. The series diverges: it has no sum to infinity

31. $32/3$

32. (a) $\$3649.96$ (b) $\$9573.44$ (c) $\$716.28$ (d) $\$2439.78$

33. 5.25%

34. 9 complete years

35. $\$10\,080$

36. $\$13\,050$

37. 720

38. 3024

39. 220

40. 315

41. 10

42. (a) $x^4 + 12x^3 + 54x^2 + 108x + 81$ (b) $1 + 14x + 84x^2$ (c) $16x^4 - 32x^3 + 24x^2 - 8x + 1$ (d) $1 - 18x + 135x^2$

43. $-720x^2$, so the coefficient is -720

44. 84

45. 5

46. $-160x^3y^3$

47. (a) $1 - x + x^2 - x^3$, valid for $|x| < 1$ (b) -2

48. (a) $1 + 2x - 2x^2$, valid for $|x| < \frac{1}{4}$ (b) $1 + \frac{x}{3} - \frac{x^2}{9}$, valid for $|x| < 1$

49. $\frac{1}{3} + \frac{x}{9} + \frac{x^2}{27}$, valid for $|x| < 3$

50. $2 + \frac{x}{12} - \frac{x^2}{288}$, valid for $|x| < 8$

51. 1.0488 (true value 1.048809)

END-OF-CHAPTER PROBLEMS

1. (a) 31 (b) 29 (c) -1

2. (a) 1 (b) 590 (c) $651 > 600$

3. (a) 3 (b) 455

4. $140.$)

5. (a) 6 (b) 2 (constant), so the sequence is arithmetic (c) 20.5

6. (a) -3 (b) 2460

7. $\frac{1}{2}$

8. (a) 4 (b) $-\frac{27}{5}$

9. (a) $\frac{1-r^n}{1-r}$ (b) $\frac{58025}{19683}$

10. (a) $\frac{3}{4}$ (b) $12\big(1-(\frac{3}{4})^n\big)$ (c) 17

11. (a) $n(2n-1)$ (b) 780

12. (a) $2x$ (b) $|2x| < 1 \Rightarrow |x| < \frac{1}{2}$, i.e. $-\frac{1}{2} < x < \frac{1}{2}$ (c) $\frac{1}{1-2x}$

13. $25 - 5\sqrt{15}$

14. -0.760. Both satisfy $|r| < 1$, so a sum to infinity exists for each

15. (a) $\frac{1}{10^6}$ (b) $\frac{1}{7}$ (c) $0.\overline{142857}$

16. (a) $\$27\,127.31$ (b) 7 years

17. (a) $\$10\,778.81$ (b) $\$9027.08$ (c) Real growth is far smaller than nominal growth

18. (a) $P(1.05)^n$ (b) $\$12\,278.26$ (c) 22.5. After **23** complete years

19. (a) $\$17\,643.78$ (b) 5.87. First below \$10 000: year **6** (c) $\$20\,136.33$

20. (a) 1260 (b) 1456

21. (a) $\frac{n(n-1)}{2}$. This equals $1+2+\cdots+(n-1)$, the Gauss pairing sum of §2.2: choosing 2 from n counts the same handshakes that the arithmetic sum counts (b) 20

22. (a) ± 2 (b) 448

23. (a) 7920 (b) 1760

24. (a) $1-2u+3u^2-4u^3+\cdots$ (b) 0.96117 (5 d.p.), matching $\frac{1}{1.02^2}$

25. (a) $2+\frac{x}{4}-\frac{x^2}{64}+\cdots$, valid for $|x| < 4$ (b) $2.04939\ldots$

26. (a) 3.43 m (b) 56.7 m

27. (a) 3 (b) No sum to infinity: $|r| = 3 > 1$

28. (a) 31 (b) 2 is constant, so v_n is geometric (c) $2^{n+1}-1$

29. (a) $4n-1$ (b) Common terms: $7, 19, 31, 43, \ldots$ Next three after 7: $\mathbf{19}, \mathbf{31}, \mathbf{43}$ (c) Common terms: $7, 19, 31, 43, \ldots$ differ by 12 (constant), so they form an AP

30. (a) $3000 + 150n$. B (compound, geometric): $3000(1.04)^n$ (b) $\$4440.73$ (c) 12

31. (a) $500(1.04)^3 + 500(1.04)^2 + 500(1.04)$ (b) $\$6\,243.18$ (c) 12.3. After **13** complete years

32. (a) The repayment M is then subtracted, leaving $L(1+i) - M$ (b) Proof (c) Proof (d) $\$386.66$ per month (e) $\$3199.36$

33. (a) $\binom{10}{k} 2^{10-k}(-3)^k x^{10-2k}$ (b) $-1\,959\,552$ (c) $-414\,720$ (d) All powers $10-2k$ are even; 3 is odd (e) 1 (f) 11 terms; $x^{10}, x^8, \ldots, x^{-8}, x^{-10}$

34. (a) Proof (b) First four terms: $3,\ 5,\ 7,\ 9$ (c) 3 (d) 440 (e) 9840

CHALLENGE QUESTIONS

35. (a) $\log_2 3$, a constant. Arithmetic (b) 332.84 (c) 19 (d) $\log_b a$, constant. Arithmetic (e) Proof (f) Equal multiplicative steps become equal distances; e.g. decibels

36. (a) $100\,500$ (b) $33\,165$ (c) $234\,168$ (d) $201\,003$, that is $S_3 + S_5 - 2S_{15}$ (e) Proof (f) $104\,104$. Verify with a GDC list, or by summing the two groups separately

37. (a) 1 (b) $16/7$ (c) $2\sqrt{3}$

Chapter 3 · Proofs

YOUR TURN

1. (a) 4 (b) Identity: true

2. (a) Proof (b) Proof (c) Proof

3. (a) Proof (b) Proof (c) Proof

4. (a) Proof (b) Proof

5. (a) -3 is an integer and is not positive (b) 2 is prime and even (c) 4 is divisible by 4 but not by 8 (d) 5, which is odd

6. (a) 1: $1 \geq 2$ is false (b) $x = 1$ (c) 2 is not odd

7. (a) Proof (b) Proof

8. (a) Proof (b) Proof

9. (a) Proof (b) Proof

10. (a) Proof (b) Proof

11. (a) Proof (b) Proof

12. (a) Proof (b) Proof (c) Proof

13. (a) Proof (b) 0 unestablished; the base case must match the range being claimed

END-OF-CHAPTER PROBLEMS

1. Proof

2. (a) Proof (b) Proof

3. 10, divisible by 5. The claim is false

4. Proof

5. (a) Proof (b) $\frac{1}{5} \times \sqrt{2}$, a non-zero rational times an irrational, so by (a) it is irrational

6. (a) Proof (b) Proof

7. Proof

8. Proof

9. Proof

10. Proof

11. Proof

12. Proof

13. Proof

14. Proof

15. Proof

16. Proof

17. Proof

18. Proof

19. Proof

20. Proof

21. (a) 2: 13 (prime) (b) 11^2, not prime

22. (a) Proof (b) Proof

23. (a) Proof (b) Proof

24. Proof

25. Proof

26. (a) Proof (b) $\frac{n(n+1)(n-1)}{3}$ (c) 2660 (d) Proof

27. (a) Proof (b) Proof (c) no contradiction follows — as it must not (d) $\sqrt{2} \times \sqrt{2} = 2$; irrationals are not closed under multiplication

28. (a) $\frac{1}{4}$ (b) $\frac{1}{n}$ (c) Proof (d) The base case is missing; nothing starts the chain (e) $n+1$ is false

CHALLENGE QUESTIONS

29. (a) Drawings give $1, 2, 4, 8, 16$ (b) Proof (c) $n + \binom{n}{2} + 2\binom{n}{4}$ (d) Proof (e) 256 (f) 30

30. (a) $65\,537$; each is prime (check divisibility by primes up to the square root) (b) Proof (c) Proof (d) $641 \times 6\,700\,417$, so F_5 is composite. One counterexample disproves the conjecture that every F_n is prime

31. (a) $\sqrt{2}$: both irrational, and a^b is rational (b) 2 is rational (c) Proof (d) This proof is non-constructive; the $\sqrt{2}$ proof is constructive

32. (a) Proof (b) Proof (c) $F_n \times F_{n+1}$; area counted two ways

33. (a) Proof (b) Proof (c) Yes; $p \mid n^p - n$ for every prime p

Chapter 4 · Lines & Quadratics

YOUR TURN

1. (a) $x + 3 \geq 0 \Rightarrow x \geq -3$ (b) $7 - x \geq 0 \Rightarrow x \leq 7$ (c) $x \neq 2$ (d) $x \neq -5$ (e) $x - 4 > 0 \Rightarrow x > 4$ *(strict: $\ln 0$ is undefined)* (f) $2x + 1 > 0 \Rightarrow x > -\frac{1}{2}$ (g) $x > 1$ (h) $x^2 - 9 \neq 0 \Rightarrow x \neq \pm 3$

2. (a) Root: $x \geq -2$; denominator: $x \neq 1$. Domain $x \geq -2$, $x \neq 1$ (b) Root: $x \geq 0$; denominator: $x \neq 9$. Domain $x \geq 0$, $x \neq 9$

3. (a) $x^2 \geq 0 \Rightarrow y \geq 1$ (b) $|x| \geq 0 \Rightarrow y \geq -3$ (c) $\sqrt{x} \geq 0$ on $x \geq 0$, so $y \geq 2$ (d) $|x+1| \geq 0$, so $3 - |x+1| \leq 3$; range $y \leq 3$

4. (a) $(x-3)^2 + 2$, so $y \geq 2$ (b) $(x-2)^2 - 4$, so $y \geq -4$ (c) Already in vertex form; $-(x-2)^2 \leq 0$, so $y \leq 5$ (d) $-(x-3)^2 + 5$, so $y \leq 5$

5. (a) -2 (b) 4 (c) 0 (d) 3

6. (a) $\frac{x+6}{3}$ (b) $2x - 1$ (c) $\frac{x-7}{2}$ (d) $\frac{1}{x} - 2$, $x \neq 0$ (e) $\frac{3x+1}{x-2}$, $x \neq 2$ (f) $\frac{2x+4}{1-x}$, $x \neq 1$

7. (a) $x \neq 1$ (b) $\frac{x+2}{x-3}$ (c) $x \neq 3$ (d) 2

8. (a) ± 2, so a vertical line meets the graph twice (b) Yes. Each x gives exactly one y

9. $\sqrt{x}$ (or $-\sqrt{x}$), with domain $x \geq 0$. Without the restriction the graph fails the horizontal line test

10. (a) 2 (b) 3 (c) -2 (d) 0; the line is horizontal

11. (a) $2x - 3$ (b) $3x - 5$ (c) $-\frac{1}{2}x + 2$ (d) $\frac{1}{2}x + 5$

12. (a) $-\frac{2}{3}x + 2$; gradient $-\frac{2}{3}$ (b) $2x - 4$; gradient 2

13. (a) $-\frac{1}{4}$ (b) $-3x + 7$, so the gradient is -3 (c) Yes, parallel (d) -1

14. (a) 2. $(3,2)$ (b) 1. $(2,1)$ (c) 1. $(3,1)$ (d) 3. $(3,2)$

15. (a) 4. $(2,4)$ (b) 2. $(3,2)$

16. (a) $8 \neq 9$, so the lines are parallel and distinct (b) None. Equal gradients, different intercepts (c) Infinitely many (d) Exactly one, at $(2,1)$

17. (a) 3. $(1,2,3)$ (b) 3. $(2,-1,3)$ (c) -2. $(3,1,-2)$ (d) 4. $(1,-2,4)$

18. (a) Last row is $(0 \quad 0 \quad a \mid b)$ (b) No solution: the first two equations describe parallel planes (c) $(4-t,\ 2,\ t)$, $t \in \mathbb{R}$ (d) $(4-2t,\ t-1,\ t)$

19. (a) $(x+2)^2-3$; vertex $(-2,-3)$ (b) $(x-2)^2-4$; vertex $(2,-4)$ (c) $(x-3)^2+2$; vertex $(3,2)$ (d) $(x+1)^2+4$; vertex $(-1,4)$

20. (a) 2 (b) 4 (c) -1 (d) 3

21. (a) -5 (b) -5 (c) -2 (d) -9; vertex $(-2,-9)$

22. (a) Maximum (b) Minimum

23. (a) 2 or 3 (b) 4 or -3 (c) -5 or 4 (d) $-\frac{1}{2}$ or 3

24. (a) -1 or -7 (b) $1 \pm \sqrt{6}$ (c) $-3 \pm \sqrt{7}$ (d) $2 \pm \frac{\sqrt{6}}{2}$

25. (a) $-1 \pm \sqrt{2}$ (b) $-3 \pm \sqrt{5}$ (c) 2 or $-\frac{1}{2}$ (d) $\frac{1}{3}$ or -1

26. (a) Factorisation (b) Completing the square, or the formula (c) The quadratic formula (d) Factorisation, as a difference of two squares: $(x-3)(x+3)$

27. (a) 0; one repeated real root (b) $-31 < 0$; no real roots (c) $16 > 0$; two distinct real roots (d) $-3 < 0$; no real roots

28. (a) 9 (b) ± 8 (c) $9 - 4k < 0 \Rightarrow k > \frac{9}{4}$ (d) 0

29. $k < -7$ or $k > 5$

30. (a) $-3 < x < 3$ (b) $(x+5)(x-3) < 0 \Rightarrow -5 < x < 3$ (c) $(x-1)(x-3) \geq 0 \Rightarrow x \leq 1$ or $x \geq 3$ (d) $-x(x-4) > 0 \Rightarrow x(x-4) < 0 \Rightarrow 0 < x < 4$

31. (a) $2x^2 - x - 6 \leq 0 \Rightarrow (2x+3)(x-2) \leq 0 \Rightarrow -\frac{3}{2} \leq x \leq 2$ (b) $x^2 - 16 \geq 0 \Rightarrow (x-4)(x+4) \geq 0 \Rightarrow x \leq -4$ or $x \geq 4$ (c) $x \in \mathbb{R}$ (d) $x^2 \geq 0$, so $x^2 + 4 \geq 4 > 0$

32. (a) $x^2 + 2x$ (b) $x^2 + 2x - 15 < 0 \Rightarrow (x+5)(x-3) < 0 \Rightarrow -5 < x < 3$. A width must be positive, so $0 < x < 3$

END-OF-CHAPTER PROBLEMS

1. (a) $\frac{x+1}{2}$ (b) Proof

2. (a) x-intercept $\frac{3}{2}$; y-intercept 3 (b) $\frac{3-x}{2}$

3. (a) $x \neq 3$ (b) $\frac{3x+1}{x-2}$

4. $2x - 7$

5. (a) $(1,6)$ (b) 2 (c) $4\sqrt{5}$

6. (a) x-intercept $(6,0)$; y-intercept $(0,4)$ (b) $-\frac{2}{3}x + 4$, gradient $-\frac{2}{3}$

7. (a) $\frac{1}{2}$ (b) $\frac{1}{2}x + \frac{3}{2}$ (c) $-2x + 9$

8. 2. $(1,2)$

9. 1. $(1,1,1)$

10. (a) $k - 7 \neq 0$, no solution) (b) $(2+t,\ 1-2t,\ t)$, $t \in \mathbb{R}$ (c) 10 the system is inconsistent: the three planes have no common point (they form a triangular prism, meeting pairwise in three parallel lines)

11. (a) $(x-2)(x+5)$ (b) $x^2 + 3x - 10$

12. $x^2 - 4x + 3$

13. (a) $(x+3)(x-1)$; roots -3 and 1 (b) -4, so vertex $(-1,-4)$ (c) Upward parabola through $(-3,0), (1,0), (0,-3)$, minimum $(-1,-4)$

14. (a) $(x-3)^2 - 4$ (b) 3 (c) $0 \Rightarrow 1$ and 5

15. $(x-4)(x+2) \leq 0 \Rightarrow -2 \leq x \leq 4$

16. No real roots $\Rightarrow \Delta < 0$: $16 - 4c < 0 \Rightarrow c > 4$

17. $(x+3)(x-2) > 0 \Rightarrow x < -3$ or $x > 2$

18. $\Delta > 0$: $k^2 - 16 > 0 \Rightarrow k < -4$ or $k > 4$

19. -2

20. $(t-3)(t-7)$, so 3 and 7

21. 5

22. Proof

23. Proof

24. 0; points $(3,9)$ and $(-1,1)$

25. (a) $-12 < 0$: no real solution (b) The line does not meet the parabola (it passes below it)

26. (a) 4 s (b) 20 m

27. 0

28. (a) $w(w+3)$ (b) 5 (reject -8)

29. 25. Maximum area 25

30. Proof

31. (a) Cubic with a local maximum then a local minimum (b) $-2.11,\ 0.254,\ 1.86$ (c) Local maximum $(-1.15,\ 4.08)$; local minimum $(1.15,\ -2.08)$ (d) $x^3 - 4x + 1$

32. (a) Proof (b) 6.4 m (c) The lorry passes (d) 12.6 m (e) The model is valid only for $|x| \leq 20$

33. (a) Proof (b) 9 (c) $1 < m < 9$ (d) The point of tangency is $(-2, 1)$ (e) $m < 1$ or $m > 9$

34. (a) $b-7$ (b) 4 (c) 7 (d) $(-1, 3, 0) + t(1, -2, 1)$, $t \in \mathbb{R}$ (e) a nonzero number, so there is no solution and the planes have no point in common

CHALLENGE QUESTIONS

35. (a) $-\frac{m^2}{4}$ (b) $\frac{m}{2}$, so they touch at $(2, 4)$ and $(-2, 4)$ (c) $-\frac{m^2}{4a}$ (d) $k < 0$: two tangents; $k = 0$: one; $k > 0$: none (e) False: a point on the parabola has one tangent

36. (a) $\frac{3-3z}{2}$ (b) Proof (c) $\lambda = 1$; infinitely many solutions (d) $k - 3 \neq 0$, which is false, so the system is inconsistent and has no solution (e) A unique solution is most likely

37. (a) Comparing (b) $S^2 - 4P \geq 0$ (c) 0. (Its roots are 5 and 1, which do sum to 6 and differ by 4.) (d) $\frac{6}{5}$ (e) Any $x^2 - Sx + 1$

Chapter 5 · Function Operations

YOUR TURN

1. (a) $2x + 3$ (b) $2x + 6$ (c) $x + 6$ (d) $4x$

2. (a) $(x-1)^2$ (b) $x^2 - 1$ (c) 4 (d) 8

3. (a) 49 (b) 9 (c) 4 (d) 14

4. (a) $\frac{1}{x+1}$ (b) $x \neq -1$ (c) $\frac{1}{x} + 1$ (d) $x \neq 0$

5. (a) $\frac{x+2}{3}$ (b) $4x - 4$ (c) $2x + 5$ (d) $\frac{1}{x} + 1$, $x \neq 0$

6. (a) $\frac{x-1}{2}$ (b) x (c) x (d) f^{-1} really is the inverse of f

7. (a) $(f \circ g)(x) = x^2 + 4x + 4$; $(g \circ f)(x) = x^2 + 2$ (b) $-\frac{1}{2}$

8. (a) Translation 3 units up (b) Translation 4 units left (c) Reflection in the x-axis (d) Translation 1 unit right

9. (a) Reflection in the y-axis (b) Reflection in the x-axis (c) Vertical stretch, factor 3 (d) Horizontal stretch, factor $\frac{1}{2}$

10. (a) Right 3, up 1: $(5, 6)$ (b) $(2, 15)$ (c) Horizontal stretch factor $\frac{1}{2}$: $(1, 5)$ (d) -3, so $(2, -3)$

11. (a) $(-1, 3)$ (b) $(-2, -1)$ (c) $(2, 3)$ (d) $(-2, \frac{3}{2})$

12. (a) Translation 3 units left (b) Vertical stretch, factor 2 (c) Translation 3 left, vertical stretch factor 2, translation 1 down (d) $\sqrt{2(x-3)}$: horizontal stretch factor $\frac{1}{2}$, then translation 3 right

13. (a) $3x^2 + 2$ (b) $3x^2 + 6$. The two differ, so the order matters

14. (a) $f(4(x-2))$ (b) Horizontal stretch factor $\frac{1}{4}$, then translation 2 right

15. (a) 3 or -1 (b) -4 or 1 (c) 3 or -2 (d) ± 3

16. (a) $-2 < x < 2$ (b) $x \leq -3$ or $x \geq 3$ (c) $x^2 - x - 6 \leq 0 \Rightarrow (x-3)(x+2) \leq 0 \Rightarrow -2 \leq x \leq 3$ (d) $x^2 - 2x - 3 > 0 \Rightarrow (x-3)(x+1) > 0 \Rightarrow x < -1$ or $x > 3$

17. (a) $3, -2$; points $(3, 9)$ and $(-2, 4)$ (b) $4, -2$; points $(4, 16)$ and $(-2, 4)$ (c) $1 \Rightarrow (1, 0)$ and $(-1, 0)$ (d) 0; points $(0, 0)$ and $(-1, 1)$

18. (a) $e^x + 1 > 0$ (b) 0.792 (from a GDC intersect or zero) (c) There is no closed-form solution

19. (a) Solution: $1 < x \leq \frac{5}{2}$ (b) Solution: $-2 < x \leq 3$

20. (a) $90\,^\circ$C (b) $58.4\,^\circ$C (c) 10.4 minutes (d) 20 is a horizontal asymptote

21. (a) $f(x)$: **even** (b) $-f(x)$: **odd** (c) $x^2 - x \neq \pm f(x)$: **neither** (d) $-f(x)$: **odd** (e) $f(x)$: **even** (f) $-f(x)$: **odd**

22. (a) f^{-1} (b) x (c) x (d) 0), so applying f twice does not return the input

23. (a) $\sqrt{x}$ (b) $x \geq 1$; $f^{-1}(x) = 1 + \sqrt{x}$ (c) $x \geq 0$, which is the range of f (d) x^3 is one-to-one on all of $\mathbb{R}$

24. (a) 5 or 1 (b) 3 or -4 (c) 2 or 4 (d) ± 5

25. (a) 2. It meets the axis there but does not cross: the graph turns at a sharp corner (b) 2, where it touches the axis (c) $(0, 4)$: the part below the axis is reflected upward (d) $f(|x|)$ receives the same input for x and $-x$

26. (a) 4 (b) -3 (c) -5 (d) None

27. (a) ± 2 (b) 0 (c) $-\frac{1}{4}$ (d) A maximum

28. (a) 3 (b) $(1, \frac{1}{2})$, a maximum (c) 0 (d) They cross

29. (a) 0; two branches, no turning point (b) Asymptotes $x = \pm 1$, $y = 0$; maximum $(0, -1)$ (c) No vertical asymptote; $y = 0$; maximum $(0, 1)$ (d) Asymptotes $x = 1$, $x = 3$, $y = 0$; maximum $(2, -1)$

30. (a) 0 (b) 1, giving $(1, -1)$ (c) 3, so $2 < x < 3$

END-OF-CHAPTER PROBLEMS

1. (a) $2x^2 - 3$ (b) $4x^2 - 12x + 9$ (c) ± 2

2. (a) $(1, 6)$ (b) $(4, 4)$ (c) $(1, 8)$ (d) Minimum at $(1, -4)$

3. (a) $(x-2)^2 + 1$ (b) Translation 2 right and 1 up (c) Minimum value 1, so range $y \geq 1$

4. $\frac{2x+1}{x-1}$

5. $x = 1$ or $x = 2$; $1 \le x \le 2$

6. (a) $-f(x)$: **odd** (b) $f(x)$: **even** (c) $x^2 - x^3 \neq \pm f(x)$: **neither**

7. (a) $\sqrt{x-2}$ (positive root) (b) Domain $x \ge 2$

8. x-intercepts ± 1, y-intercept 1, peak $(0,1)$

9. $-\frac{1}{2}$

10. $(0,1) \mapsto (-1,-1)$; $(2,0) \mapsto (1,-2)$; $(3,4) \mapsto (2,2)$

11. (a) $\frac{x-2}{3}$ (b) -1

12. (a) Domain $x \ge 1$; range $y \ge 0$ (b) x^2+1, domain $x \ge 0$

13. (a) 0 (b) $3 + \frac{2}{x}$

14. Reflection in the x-axis, then 1 right and 2 up; vertex $(1,2)$

15. -1. (The point equidistant from 2 and -4.)

16. $-5 < 2x - 1 < 5 \Rightarrow -4 < 2x < 6 \Rightarrow -2 < x < 3$

17. (a) $-f(x)$: **odd** (b) $f(x)$: **even** (c) $-x+1 \neq \pm f(x)$: **neither**

18. 5

19. (a) $(x-3)^2 - 9$ (b) Translation 3 right, 9 down; vertex $(3,-9)$

20. Only intersection: $(1,1)$

21. (a) Domain $x \neq -2$; range $y \neq 0$ (b) 0

22. (a) 2 (b) $\frac{x-2}{3}$

23. (a) 3 (b) Translation 2 right and 1 down

24. (a) $x(x-2)$: roots 0 and 2; vertex $(1,-1)$ (b) Roots 0 and 2; y-intercept 0; maximum $(1,1)$

25. Solution: $-2 < x \le 3$

26. (a) -1), so f is one-to-one (b) $-1 + \sqrt{x+4}$, domain $x \ge -4$ (the range of f)

27. (a) $\frac{7}{2}$.) (b) $|x-2| \ge |3x+1| \iff (x-2)^2 \ge (3x+1)^2 \iff 8x^2+10x-3 \le 0$, so $-\frac{3}{2} \le x \le \frac{1}{4}$

28. $\frac{x}{1+2x}$

29. Proof

30. 0

31. Proof

32. Proof

33. (a) Vertex $(2,-1)$ (b) y-intercept $(0,3)$ (c) y-intercept $(0,3)$ marked (d) ± 3, minima at $(\pm 2,-1)$, y-intercept $(0,3)$ (e) $x^2 - 8x + 18$

34. (a) the range is $y \in \mathbb{R}$, $y \neq 3$ (the horizontal asymptote) (b) Domain: $x \neq 3$ (the range of f) (c) $\frac{3x+5}{x+4}$ (d) Domain: $x \in \mathbb{R}$, $x \neq -4$ (e) 2

35. (a) Correct shape (b) 2 (c) $(0, -\frac{1}{4})$, a maximum (d) 0 (e) 0 shown

CHALLENGE QUESTIONS

36. (a) x (b) $\frac{(a^2+bc)x+b(a+d)}{c(a+d)x+(bc+d^2)}$ on collecting terms (c) Proof (d) A constant function has no inverse (e) Proof

37. (a) Equal (b) Equal (c) Proof (d) Transformations commute exactly when they act on different variables

38. (a) x-intercepts $(-2,0)$ and $(2,0)$; local maximum $(0,4)$ (b) 10 only, **2** solutions (c) No other k has this property (d) 4: 3 solutions. $k > 4$: 2 solutions (e) $-m$: 3; $k > -m$: 2

Chapter 6 · Rational & Exponential Functions

YOUR TURN

1. (a) 3 (b) 5 (c) -2 (d) -1

2. (a) 2 (ratio of leading coefficients) (b) 1 (c) 3 (d) 0: the numerator has lower degree than the denominator

3. (a) 1; x-intercept $(1,0)$; y-intercept $(0,-\frac{1}{2})$ (b) 2; x-intercept $(-\frac{3}{2},0)$; y-intercept $(0,-3)$

4. (a) Domain $x \neq 5$; range $y \neq 0$ (b) Domain $x \neq 0$; range $y \neq 4$

5. (a) $3 - \frac{9}{x+1}$ (b) 3, so they cross at $(-1,3)$ (c) Two branches centred on $(-1,3)$; intercepts $(2,0)$ and $(0,-6)$

6. (a) 2 (b) -3 (c) -3 (d) -2

7. (a) x (b) $x+1$ (c) $x+2$ (d) $x+3$

8. (a) 0 (b) 1 (c) $x-2$

9. (a) 0 (b) x (c) 0, which has no solution (d) x

10. (a) 8 (b) $\frac{1}{4}$ (c) 1 (d) 3

11. (a) 0; y-intercept 1 (b) 0; y-intercept 5 (c) 5; y-intercept 6 (d) 3

12. (a) Decay: the base 0.5 lies between 0 and 1 (b) Growth: the base $3 > 1$ (c) Decay: the exponent has a negative coefficient, $-0.2 < 0$ (d) $\left(\frac{1}{2}\right)^x$, and $\frac{1}{2} < 1$

13. (a) 3 (b) 0 (c) 2 (d) 0.415

14. (a) $e^{x \ln 2}$ (b) $e^{x \ln 5}$

15. (a) 800 (b) 1080 (c) 13.5 years

16. (a) 15.1 (b) 3.47 days

17. (a) 3 (b) 0 (c) 1 (d) 3

18. (a) 1 (b) 9 (c) 20.1 (d) 17

19. (a) $x > 2$ (b) $x > -5$ (c) $2x - 6 > 0 \Rightarrow x > 3$ (d) $4 - x > 0 \Rightarrow x < 4$

20. (a) 0; x-intercept $(1, 0)$ (b) 1 $\Rightarrow$ intercept $(0, 0)$ (c) 1 $\Rightarrow$ intercept $(3, 0)$ (d) 1 $\Rightarrow$ intercept $(-4, 0)$

21. (a) $\ln(x - 2)$, domain $x > 2$ (the range of f) (b) $e^x + 1$, domain all real x (c) $\frac{1}{2}\ln(\frac{x-1}{3})$, domain $x > 1$

22. (a) $9 - x^2 > 0$ (b) $-3 < x < 3$

23. (a) $-\frac{1}{x-1} + \frac{1}{x-2}$ (b) $\frac{1}{x-2} - \frac{1}{x+3}$ (c) $\frac{1}{x+1} - \frac{1}{x+4}$ (d) $\frac{1}{2(x-1)} + \frac{1}{2(x+1)}$

24. (a) $\frac{1}{x} + \frac{1}{x+1}$ (b) $-\frac{2}{x} + \frac{3}{x-2}$ (c) $\frac{3}{2(x-3)} + \frac{1}{2(x+1)}$ (d) $\frac{11}{3}$. $\frac{4}{3(x-1)} + \frac{11}{3(x+2)}$

25. (a) $(x - 2)(x + 2)$; then $\frac{2}{x-2} + \frac{2}{x+2}$ (b) $(x - 1)(x + 2)$; then $\frac{8}{3(x-1)} - \frac{5}{3(x+2)}$

26. (a) Numerator and denominator both have degree 2 (b) $1 + \frac{1}{x^2-1}$ (c) $\frac{1}{2(x-1)} - \frac{1}{2(x+1)}$, so the whole fraction is $1 + \frac{1}{2(x-1)} - \frac{1}{2(x+1)}$

END-OF-CHAPTER PROBLEMS

1. (a) 2 (b) x-intercept $(-\frac{3}{2}, 0)$; y-intercept $(0, -3)$ (c) Reciprocal curve centred on $(1, 2)$, two branches

2. (a) $x \neq -3$ (b) 1 (c) $\frac{3x+2}{1-x}$

3. (a) 0 (b) x (c) ± 1

4. 2

5. $\frac{5}{2}$

6. (a) $\ln x$ (b) $x > 0$

7. 1

8. $\frac{1}{2}\ln 5$

9. 5. $\frac{2}{x-1} + \frac{5}{x+2}$

10. (a) 3 (b) Proof

11. (a) 500 (b) 4000 (c) 4 hours

12. (a) 80 g (b) Half-life 5 days (the exponent is $t/5$) (c) 20 g

13. -2, as log needs $x > 2$)

14. $-\frac{1}{4}$. $\frac{1}{4(x-2)} - \frac{1}{4(x+2)}$

15. 2

16. (a) \$20000 (b) \$10440 (c) 0)

17. (a) $x + 2$ (b) $x \neq 1$ (c) $x + 2$ with a hole (open point) at $(1, 3)$

18. 1.58

19. 0.176

20. -4. $\frac{5}{x} - \frac{4}{x+1}$

21. (a) 2 (b) $-\frac{1}{2}$ (c) -1

22. $\ln 2$

23. 2^{-x} rises to the left (mirror images in the y-axis)

24. 3

25. Proof

26. 2

27. $-\frac{1}{x} + \frac{1}{x-1} + \frac{1}{x+1}$

28. 3

29. (a) 2 (b) -3

30. (a) Proof (b) $x - 2$ (c) Proof (d) Proof (e) Both asymptotes drawn and labelled; no x-intercept shown

31. (a) 100 people (b) 2635 people (c) 4.86 days (d) 5000: the greatest number who could hear the rumour (e) Proof

32. (a) $x > 2$ (b) $\frac{5}{2}$ (c) 2 (d) Range of f^{-1} is $y > 2$ (the domain of f) (e) Translation 2 right, then $\ln 2$ up

CHALLENGE QUESTIONS

33. (a) Proof (b) Proof (c) 1 is a horizontal asymptote of the graph of the partial sums (d) $\frac{3}{4}$ (e) Proof

34. (a) Proof (b) $\frac{\ln 2}{k}$, independent of where the halving starts (c) 12.4 minutes (d) 22.4, so 22 minutes to the nearest minute (e) 20 is a horizontal asymptote of the graph. (Accept: the model is idealised; a real cup does reach room temperature.)

35. (a) $\frac{cd^2-bde+ae^2}{d^2}$ (b) $y = \frac{a}{d}x + \frac{bd-ae}{d^2}$ (c) ae (d) Proof (e) $x + 4$ with a hole at $(1, 5)$

Chapter 7 · Polynomials

YOUR TURN

1. (a) Degree 4; lead coefficient 3 (b) Degree 3; lead coefficient -1 (c) Degree 3; lead coefficient 1 (d) Degree 5; lead coefficient -2

2. (a) $x \to +\infty$: $y \to +\infty$; $x \to -\infty$: $y \to -\infty$ (b) Down at both ends: even degree, negative lead coefficient (c) $x \to +\infty$: $y \to +\infty$; $x \to -\infty$: $y \to -\infty$ (d) Even degree, negative lead coefficient: down at both ends

3. (a) 5 (b) 3 (c) 0 or 2 (d) Odd degree gives opposite end behaviours on a continuous curve

4. (a) $2,\ -3,\ 5$; the graph crosses at each (b) -4 (crosses) (c) 3 (touches) (d) 2, and the graph touches at both: each factor is squared

5. (a) 3 with multiplicity 2 (b) 9, so $(0, 9)$ (c) Leading term x^3: down on the left, up on the right (d) 3, passes through $(0, 9)$, rising to the right

6. (a) $a(x-2)^2(x+1)$ (b) 1 (c) $(x-2)^2(x+1)$

7. (a) $4x^3 - x^2 + 2x - 4$ (b) $x^4 - 2x^3 + 4x^2 - 3$ (c) $x^4 + x^3 + 7x - 3$ (the x^2 terms cancel) (d) $x^3 - 8$

8. (a) Quotient $x+3$, remainder 0 (b) Quotient $x^2 - 2x$, remainder 0 (c) Quotient $x^2 + 3x + 2$, remainder 3 (d) Quotient $x^2 + 2x + 4$, remainder 0

9. (a) Write $2x^3 - x^2 + 0x - 3$: quotient $2x^2 - 3x + 3$, remainder -6 (b) Write $x^3 + 3x^2 + 0x - 4$: quotient $x^2 + x - 2$, remainder 0 (c) Write $2x^4 - 3x^3 + 0x^2 + x - 6$: quotient $2x^3 + x^2 + 2x + 5$, remainder 4

10. (a) Quotient $x^2 + 1$, remainder 0 (b) Quotient $x^2 - 1$, remainder 0

11. (a) 2 (b) 0 (c) 2 (d) -5

12. (a) 1 (b) -2

13. (a) 3 is a root (b) 7 (c) $(x+1)$ (d) 0, so $(x-1)$ is a factor

14. (a) -5 (b) -1

15. (a) 0: yes (b) 0: yes (c) $4 \neq 0$: no (d) 0: yes

16. (a) $x(x-1)(x+1)$ (b) $x(x-2)(x+2)$ (c) $(x-1)(x+1)(x+2)$ (d) $(x-1)(2x-1)(x+2)$

17. (a) $(x-3)$ (b) $(x-2)^2(x+1)$. The factor $(x-2)$ appears twice

18. (a) $1,\ -2,\ 4$ (b) -1 (c) $0,\ 2,\ -2$ (d) $0,\ -2,\ 1$

19. (a) $1,\ 2,\ 3$ (b) $1,\ 2,\ -1$ (c) $1,\ \frac{1}{2},\ -2$

20. (a) $\pm 1,\ \pm 2$ (b) $-3 < 0$, so it contributes no real root

21. (a) Sum 7; product 10 (b) Sum $-\frac{3}{2}$; product $-\frac{5}{2}$ (c) Sum -4; product 4 (d) Sum $\frac{2}{3}$; product $-\frac{8}{3}$

22. (a) 6 (b) $\frac{3}{2}$ (c) -5

23. (a) 0 (b) -7 (c) ± 6 (d) 0

24. (a) 3 (b) $\frac{7}{6}$ (c) $\frac{25}{4}$ (d) $\frac{1}{4}$

25. (a) 3 (b) -3 (c) 0

26. (a) -6 (b) 1 (c) 14

END-OF-CHAPTER PROBLEMS

1. (a) Proof (b) $(x-1)(x-3)(x+2)$ (c) $1,\ 3,\ -2$
2. (a) Proof (b) $(x-2)(2x+1)(x+3)$
3. 1
4. 6
5. Cubic; roots $x = -1,\ 1,\ 3$; y-intercept $(0, 3)$
6. q
7. (a) $(x-1)(x+1)(x+2)$ (b) $1,\ -1,\ -2$
8. $1, 2, -3$
9. 19
10. 5
11. (a) $x^3 - 6x^2 + 11x - 6$ (b) Proof
12. (a) $\pm 1,\ \pm 2$ (b) Four
13. $-2,\ -3$
14. ± 6
15. $-\frac{1}{2}$
16. (a) Proof (b) $-2,\ -\frac{1}{2},\ 3$
17. 18
18. $1,\ -\frac{1}{2},\ -2$
19. 8
20. 8
21. (a) 5 (b) $-1 \pm \sqrt{2}$
22. 3 (crosses); cubic rising to the right
23. 6. So $x^2 - 6x + 6$
24. (a) Proof (b) 0 has no real solution, so there are no other real roots
25. 1, so the remainder is $2x + 1$
26. 14
27. Proof
28. Proof

29. $-k$ lies strictly between them: $-2 < -k < 2$, i.e. $-2 < k < 2$

30. (a) Proof (b) All three factors correct (c) All three (d) As $x \to -\infty$, $f(x) \to -\infty$ (e) y-intercept $(0, 6)$ marked (f) Both parts, with the endpoints included

31. (a) Proof (b) 3 (c) $(x+1)(x^2-3x+6)$ (d) The quadratic factor has negative discriminant, so it contributes no real roots (e) 4: the remainder is $4x+4$

32. (a) $-r$ (b) 30 (c) $-\frac{3}{10}$ (d) $6.$) (e) $0.$ (Roots $-2,\ 4,\ 10.$)

CHALLENGE QUESTIONS

33. (a) p (b) $p^2 - 2q$ (c) $p(p^2 - 3q)$ (d) Proof (e) 47 (f) The recurrence builds each S_k from integers

34. (a) The constant term a_0 equals the lead coefficient a_n, and $a_n \neq 0$ by definition of degree (b) Proof (c) Proof (d) $\frac{-3-\sqrt{5}}{2}$. The last two multiply to 1, as part (b) requires (e) Proof

Chapter 8 · Solids, Circles & Triangles

YOUR TURN

1. (a) 5 (b) 7 (c) 5 (d) 5

2. (a) $(3, 3, 2)$ (b) $(4, 1, 2)$

3. (a) 25 (b) 16 (c) 36 (d) 7

4. (a) 5 (b) 13 (c) $67.4°$

5. (a) 36π (b) 12π (c) 60 (d) 160π

6. (a) 100π (b) 65π (c) 16π: total 48π (d) 80π

7. (a) 10 (b) 60π

8. (a) The cone's curved surface and the hemisphere's curved surface (b) 18π. Total 33π (c) 18π. Total 30π

9. (a) 9.43 (b) 12.9 (c) 12.2 (d) $43.5°$

10. (a) 7 (b) 12.4 (c) $46.6°$ (d) $78.5°$

11. (a) 10 (b) 27 (c) 20.2

12. (a) $65.4°$ or $114.6°$ (b) $159.6 < 180$. Both leave a positive third angle, so both triangles exist (c) SAS fixes the triangle uniquely; no ambiguous sine arises

13. (a) $22.6°$ (b) $26.6°$

14. (a) 172 m (b) 35.8 m (c) 165 m

15. (a) $270°$ (b) $300°$ (c) $245°$ (d) $020°$; adding would give the invalid bearing $380°$

16. (a) 5.64 m (b) 2.05 m

17. (a) $90°$, so the ship turns through a right angle at Q (b) 64.0 km (c) $081°$

18. (a) $\frac{\pi}{3}$ (b) $\frac{3\pi}{4}$ (c) $\frac{7\pi}{6}$ (d) $\frac{\pi}{2}$

19. (a) $135°$ (b) $120°$ (c) $57.3°$

20. (a) 48 (b) 18.8 (c) 97.2 (d) 12.6

21. (a) 2.5 rad (b) 18

22. (a) 64 (b) 29.1 (c) 34.9 (d) 32

END-OF-CHAPTER PROBLEMS

1. (a) 14.1 (b) $45°$

2. (a) $4\sqrt{2}$ (b) $78.5°$

3. (a) 13 (b) 100π (c) 90π

4. 87π

5. (a) $103.1°$ (b) 42.9

6. (a) 12.7 (b) 34.9

7. (a) 7.71 (b) 34.7

8. (a) Proof (b) 64.0 km (c) $081°$

9. (a) 14.4 cm (b) 86.4 cm^2 (c) 38.4 cm

10. (a) 1.67 rad (b) 10 cm

11. (a) 75 (b) 49.9 (c) 25.1

12. (a) $63.4°$ or $116.6°$ (b) 4.87

13. 166

14. (a) 12 (b) 13 (c) 65π

15. 54.6 m

16. (a) 400 (b) 13 (c) 360

17. $36.9°$ or $143.1°$

18. 28.8 km

19. (a) 9 (b) Midpoint $(2, 2, 3.5)$ (c) Proof

20. 0.4 m

21. (a) $\frac{\pi}{6}$; (b) $\frac{5\pi}{4}$; (c) $\frac{5\pi}{3}$

22. 106 m

23. $49.9°$ or $130.1°$

24. 23.6

25. 11.7

26. 0.5

27. 16.3

28. 22.5

29. (a) Proof (b) 32.8 km (3 s.f.) (c) $080.9°$, i.e. $081°$ (d) 223 km^2 (3 s.f.) (e) 75.8 km

30. (a) 10 cm (b) 11.2 cm) (c) $26.6°$ (d) $44.3°$ (e) 236 cm^2

31. (a) 565 cm^3) (b) 4 cm (c) 168 cm^3 (d) $\frac{8}{27}$, not $\frac{2}{3}$ (e) 11.9 cm (3 s.f.)

CHALLENGE QUESTIONS

32. (a) Proof (b) $12 > b$: **one** triangle (c) CB must reach from C to that line, so $a < h$ is impossible and no triangle exists (d) Exactly one triangle; it is right-angled at B (e) $180°$ and no third angle is left: it is degenerate, and only one triangle exists (f) None for $a \le b$; exactly one for $a > b$

33. (a) Proof (b) Proof (c) Proof (d) $294°$ (e) $\frac{\sqrt{2}}{2}$

34. (a) Proof (b) 1.333×10^7 cubits (c) A change of only one arcminute moves the answer by about 6% (d) Proof (e) Quote the result to at most two significant figures

Chapter 9 · Trigonometric Functions

YOUR TURN

1. (a) $\frac{1}{2}$ (b) $\frac{\sqrt{2}}{2}$ (c) $\sqrt{3}$ (d) 1

2. (a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) $-\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{2}}{2}$

3. (a) First and second (b) Second and third (c) First and third: sin and cos share a sign there (d) Fourth only

4. (a) $P(\cos\theta,\ \sin\theta)$ (b) 1^2 (c) 0, so the quotient divides by zero. Geometrically the line OP is vertical and has no gradient

5. (a) 1 (b) $\sin 2\theta$ (c) $\cos 2\theta$ (d) $\cos 2\theta$

6. (a) $\frac{4}{5}$ (b) $\frac{3}{4}$ (c) $\frac{24}{25}$ (d) $\frac{7}{25}$

7. (a) $\frac{4}{5}$, positive in the second quadrant (b) $-\frac{4}{3}$ (c) $-\frac{24}{25}$ (d) $-\frac{7}{25}$

8. (a) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (b) $\frac{\sqrt{6}+\sqrt{2}}{4}$

9. (a) Proof (b) Proof

10. (a) 0 (b) 0 (c) -2 (d) 3

11. (a) $-1 \le y \le 3$ (b) $-4 \le y \le 2$ (c) Amplitude 2, midline -1: $-3 \le y \le 1$

12. (a) π (b) $\frac{\pi}{2} + n\pi$) (c) $\frac{\sin x}{\cos x}$, so it is undefined

13. (a) 22 (b) $\frac{\pi}{10}$ (c) 2 (the bottom) (d) 22 m, the height of the axle

14. (a) 2 (b) 1 (c) 1 (d) -2

15. (a) $\sec^2\theta$ (b) $\operatorname{cosec}^2\theta$ (c) $\tan\theta$ (d) 1

16. (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{2}$ (c) $-\frac{\pi}{4}$ (d) $\frac{2\pi}{3}$

17. (a) arcsin: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$. arccos: $0 \le y \le \pi$ (b) $\frac{\pi}{4}$ (c) $\frac{2\pi}{3}$

18. (a) $\frac{\pi}{6},\ \frac{5\pi}{6}$ (b) $\frac{\pi}{2},\ \frac{3\pi}{2}$ (c) $\frac{\pi}{4},\ \frac{5\pi}{4}$ (d) $0,\ 2\pi$

19. (a) $\frac{7\pi}{6},\ \frac{11\pi}{6}$ (b) $\frac{\pi}{6},\ \frac{11\pi}{6}$ (c) $\frac{5\pi}{4},\ \frac{7\pi}{4}$ (d) $\frac{2\pi}{3},\ \frac{5\pi}{3}$

20. (a) $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$ (b) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $\frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}$

21. (a) $0,\ \pi,\ 2\pi,\ \frac{\pi}{6},\ \frac{5\pi}{6}$ (b) $\frac{\pi}{3},\ \frac{5\pi}{3},\ \pi$ (c) $\frac{\pi}{2}$

END-OF-CHAPTER PROBLEMS

1. (a) $\frac{\sqrt{3}}{2}$ (b) $-\frac{1}{2}$ (c) $-\sqrt{3}$ (d) $-\frac{\sqrt{2}}{2}$ (e) $-\frac{1}{2}$ (f) -1

2. (a) $\left(-\frac{\sqrt{3}}{2},\ \frac{1}{2}\right)$ (b) $\frac{5\pi}{4}$ (c) $\frac{\sin}{\cos}$ is a quotient of two negatives, so $\tan\theta > 0$ (d) $\frac{\sqrt{3}}{2}$ cuts the circle at two points, one in each of the quadrants

3. (a) Proof (b) Proof (c) Proof (d) Proof (e) Proof (f) Proof (g) Proof

4. (a) $\frac{4}{5}$ (positive in the second quadrant) (b) $-\frac{4}{3}$ (c) $-\frac{24}{25}$ (d) $-\frac{7}{25}$

5. (a) $\frac{1}{\sqrt{5}}$ (b) $\frac{4}{5}$ (c) $-\frac{3}{5}$

6. (a) $2+\sqrt{3}$ (b) $\frac{\sqrt{6}+\sqrt{2}}{4}$

7. (a) Proof (b) Proof (c) Proof

8. (a) $\frac{\pi}{6},\ \frac{5\pi}{6}$ (b) $\frac{\pi}{6},\ \frac{7\pi}{6}$ (c) $\frac{\pi}{4},\ \frac{3\pi}{4},\ \frac{5\pi}{4},\ \frac{7\pi}{4}$ (d) $\frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{5\pi}{3}$

9. (a) $\frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{3\pi}{2}$ (b) $\frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{5\pi}{3}$ (c) $0, 2\pi$ (d) $\frac{\pi}{3},\ \frac{5\pi}{3}$

10. (a) $\frac{\pi}{12},\ \frac{5\pi}{12},\ \frac{13\pi}{12},\ \frac{17\pi}{12}$ (b) $\frac{\pi}{2}$ (c) $0,\ \frac{\pi}{2},\ 2\pi$

11. $\frac{\pi}{3}$

12. (a) $\frac{\pi}{4}$ (b) $-\frac{\pi}{6}$ (c) π (d) $\frac{\pi}{6}$, not $\frac{5\pi}{6}$, which lies outside the range of arcsin

13. (a) π (b) Maximum 4, minimum -2

14. 2

15. Maximum 0 at $x = 0,\ 2\pi$; minimum -2 at $x = \pi$; y-intercept 0

16. (a) 2 m (b) 12 hours, the time from one high tide to the next

17. (a) 18 m (b) 30 seconds

18. (a) 0.927 (b) Maximum value 5

19. $M = 13$ at $x \approx 0.395$

20. (a) $\sin\theta$, so the value is 0.565 (b) 0.362 as well (c) Adding 2π returns P to the same point (d) 0

CHALLENGE QUESTIONS

21. (a) Proof (b) Proof (c) $25\sin(\theta + 1.287)$; maximum 25, minimum -25 (d) Proof (e) 1 happens once per period, and the shifted interval contains exactly one such point, so there is exactly one solution

22. (a) Proof (b) $4c^3 - 3c$ (c) $16c^5 - 20c^3 + 5c$ (d) 2^0), the pattern follows by induction (e) $\sin n\theta$ is odd in θ; polynomials in $\cos\theta$ are even

23. (a) $\frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}$ — **6** (b) $b > 0$: u increases from 0 to $2\pi b$ (c) $2b$ (d) The count drops from $2b$ to b to 0 as $|k|$ passes through 1 (e) 1 for cosine

Chapter 10 · Vectors

YOUR TURN

1. (a) 5 (b) 3 (c) 5 (d) $\sqrt{3}$

2. (a) $\begin{pmatrix}1\\1\\7\end{pmatrix}$ (b) $\begin{pmatrix}1\\3\\-1\end{pmatrix}$ (c) $\begin{pmatrix}1\\0\\11\end{pmatrix}$ (d) $\begin{pmatrix}3\\7\\5\end{pmatrix}$

3. (a) $\begin{pmatrix}0\\3/5\\4/5\end{pmatrix}$ (b) $\begin{pmatrix}2/3\\-1/3\\2/3\end{pmatrix}$ (c) $\begin{pmatrix}2\\4\\4\end{pmatrix}$ (d) 1. The direction is unchanged

4. (a) $2\begin{pmatrix}1\\2\\3\end{pmatrix}$ (b) $6 \neq 5$, so no single scalar works

5. (a) $\begin{pmatrix}3\\-2\\4\end{pmatrix}$ (b) $\sqrt{29}$ (c) $(\frac{5}{2},\ 1,\ 1)$ (d) $\begin{pmatrix}-3\\2\\-4\end{pmatrix}$, the negative of $\overrightarrow{AB}$

6. (a) $(4, 0, 3)$ (b) $(6, 9, 3)$ (c) Proof

7. (a) $\mathbf{a} + \mathbf{b}$ (b) Midpoint of AB: $\frac{1}{2}(\mathbf{a} + \mathbf{b})$ (c) Proof (d) Proof

8. (a) 8 (b) 0 (c) 0 (d) 0

9. (a) 45° (b) 27.3° (c) 60°

10. (a) 6 (b) 8 (c) 0 (d) The component form agrees: $a_1^2 + a_2^2 + a_3^2$ is exactly $|\mathbf{a}|^2$

11. (a) $\mathbf{a}+\mathbf{b}$ and $\mathbf{a}-\mathbf{b}$ (b) $|\mathbf{a}|^2 - |\mathbf{b}|^2$ (c) Proof (d) 0

12. (a) $\begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\-1\end{pmatrix}$ (b) $\begin{pmatrix}3\\3\\3\end{pmatrix}$, or the simpler $\begin{pmatrix}1\\1\\1\end{pmatrix}$ (c) $\begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\1\end{pmatrix}$

13. (a) 2λ (b) $\frac{z-3}{4}$ (c) All three agree

14. (a) 250 km h^{-1} (b) $\begin{pmatrix}10\\5\end{pmatrix} + t\begin{pmatrix}150\\200\end{pmatrix}$ (c) $\begin{pmatrix}310\\405\end{pmatrix}$

15. (a) $\begin{pmatrix}0\\0\\1\end{pmatrix}$ (b) $\begin{pmatrix}0\\0\\6\end{pmatrix}$ (c) $\begin{pmatrix}4\\-2\\1\end{pmatrix}$ (d) $\begin{pmatrix}-3\\6\\-3\end{pmatrix}$

16. (a) 7.5 (b) $\begin{pmatrix}0\\0\\2\end{pmatrix}$, so the area is 2 (c) 2.29 (d) 0. A vector of zero magnitude is the zero vector

17. (a) 0 (b) $\begin{pmatrix}2\\-1\\3\end{pmatrix}$ (or any scalar multiple) (c) d (d) 3 , so yes

18. (a) 6 (b) 80.4°

19. (a) 2, giving $(5, 2, 0)$ (b) $\begin{pmatrix}2\\1\\-3\end{pmatrix}$ (c) 11.4°

20. (a) 3 (b) $\sqrt{5}$ (c) 3 (d) 3 , the same as before

21. (a) 3 (b) $\frac{14}{3}$ (c) 3 (d) 3, from part (a)

22. (a) No solution, so the lines do not meet: they are skew (b) 2 (c) $\sqrt{2}$ (d) 0: the lines intersect at $(2, 3, 3)$

END-OF-CHAPTER PROBLEMS

1. (a) $\begin{pmatrix}4\\5\\0\end{pmatrix}$ (b) 3 (c) $\begin{pmatrix}2/3\\-1/3\\2/3\end{pmatrix}$

2. (a) $\begin{pmatrix}2\\3\\2\end{pmatrix}$ (b) $\sqrt{17}$ (c) $(2, \frac{1}{2}, 3)$

3. $-\frac{1}{2}$

4. 60°

5. (a) $(5, 2, 3)$ (b) 2, so P lies on l

6. 60.5°

7. $3\sqrt{5}$

8. $\frac{\sqrt{3}}{2}$

9. (a) -3 (b) $3 \neq -3$, so B does not lie on Π

10. 1

11. (a) 5 (b) $\begin{pmatrix}4/5\\-3/5\\0\end{pmatrix}$ (c) $\begin{pmatrix}8\\-6\\0\end{pmatrix}$

12. $(6, 9, 3)$

13. $(4, -1, 4)$

14. 27.3°

15. Proof

16. 53.1°

17. 2 . Intersection $(3, 2, 1)$

18. (a) $\begin{pmatrix}2\\1\\-2\end{pmatrix}$ (b) $(\frac{4}{3}, \frac{2}{3}, -\frac{4}{3})$

19. (a) 0 (b) 90°

20. 6

21. $\begin{pmatrix}2\\-1\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\2\end{pmatrix}$

22. 70.5°

23. 25

24. -2

25. (a) $\begin{pmatrix}2\\-1\\10\end{pmatrix} + t\begin{pmatrix}3\\4\\0\end{pmatrix}$ (b) 5 m s^{-1} (c) 5 s

26. $\begin{pmatrix}1\\1\\0\end{pmatrix} + t\begin{pmatrix}0\\1\\1\end{pmatrix}$

27. (a) $\begin{pmatrix}2\\1\\0\end{pmatrix} + t\begin{pmatrix}1\\0\\-1\end{pmatrix}$ (b) 2 (c) A triangular prism; the planes have no common point

28. 3 . Consistent, so the lines intersect at $(2, 3, 3)$

29. 2

30. Unit vectors: $\pm\frac{1}{3\sqrt{5}}\begin{pmatrix}-2\\5\\-4\end{pmatrix}$

31. (a) 6 (b) $\frac{7}{2}$

32. 26.4°

33. (a) $\begin{pmatrix}2\\1\\-1\end{pmatrix}$ (b) Correct signs throughout (c) 4.06) (d) Check (e) 1.85 (f) $\frac{5}{2}$

34. (a) Proof (b) Point of intersection $\left(\frac{11}{3}, \frac{2}{3}, \frac{13}{3}\right)$ (c) 90° (d) 5 (e) The lines intersect; skew lines lie in no plane

35. (a) 500 km h^{-1} (b) $\begin{pmatrix}600-300t\\100t\\0\end{pmatrix}$ (c) Proof (d) 190 km

CHALLENGE QUESTIONS

36. (a) The perpendicular is the shortest segment from P to l (b) Proof (c) 1.53 (d) 1.53, the same answer (e) Proof

37. (a) 3, which is false. No point lies on both, so the lines are skew (b) $\mathbf{d}_1 \times \mathbf{d}_2$ is the only direction perpendicular to both (c) Proof (d) 0.535 (e) correct: intersecting lines are zero distance apart

38. (a) 9 (b) Proof (c) The magnitude, and so the volume, is the same in every case (d) The triple product is zero, so the three vectors are coplanar (e) $\frac{3}{2}$

Chapter 11 · Statistics

YOUR TURN

1. (a) Discrete: it can only take whole-number values (b) Continuous: it can take any value in a range (c) Discrete (d) Continuous

2. (a) 20 males (b) 30 females (c) Each group appears in proportion to the population

3. (a) 50 (b) Systematic sampling (c) Gym users exercise more than the general population (d) 40

4. (a) 10 (b) 9 (c) 8 (d) 10

5. (a) 10.5 (b) 7, which occurs three times (c) 9 (d) 16.5

6. (a) 12 (b) 21

7. (a) 22 (b) 63 (c) 2.86 (d) 3, the value with the highest frequency

8. (a) 5, 15, 25, 35 (b) 19.3 (c) $10 \leq x < 20$ (frequency 10) (d) Each value is assumed to lie at its class midpoint

9. (a) 18 (b) 5 (c) 14 (d) 9

10. (a) 3.06 (b) 8.82 (c) 3.37

11. (a) 18 (b) 40; standard deviation unchanged at 5 (c) 6 (d) 18. The -5 does not affect the spread

12. (a) All distances between values are unchanged (b) The IQR ignores the extreme quarters; the range does not

13. (a) 9, so the median is the 5th value: 10 (b) 14 (c) 4, 6.5, 10, 14, 18 (d) 7.5

14. (a) 80 (b) 0 (c) 34

15. (a) 12 (b) 20, 40, 60 (c) Histograms show continuous data; bar charts show separate categories

16. (a) Strong negative linear correlation (b) Very weak, or effectively no, linear correlation (c) Very strong positive linear correlation (d) Moderate negative linear correlation

17. (a) 0.936 (b) 0.989

18. (a) 0 (b) Correlation shows association, not cause

19. (a) $1.7x + 0.9$ (b) $1.45x + 2.3$

20. (a) 19 (b) 18 (c) 40 lies far outside the range $2 \leq x \leq 15$ used to fit the line. There is no evidence the relationship continues that far, so the prediction is extrapolation and unreliable

21. (a) $1.5x + 8$ (b) The x on y line

END-OF-CHAPTER PROBLEMS

1. (a) 23.25 min (b) $20 \leq t < 30$ (c) $20 \leq t < 30$

2. (a) Five-number summary: 12, 18.5, 23, 27.5, 46 (b) 41 (c) Positively skewed: a long right tail, with the maximum far above Q_3

3. (a) 0.991. A very strong positive linear correlation: more revision is associated with higher scores (b) $2.30x + 5.62$ (3 s.f.) (c) 0 lies below the data range) (d) 21.7, so about 22 (e) 25 is far outside the data range ($2 \leq x \leq 12$), so this would be unreliable extrapolation

4. (a) 8 (b) 12 (c) Adding a constant shifts the centre but not the spread

5. (a) 30 (b) Each town is represented in proportion to its size

6. (a) Plot $(10, 4), (20, 16), (30, 34), (40, 52), (50, 60)$ and join with a smooth increasing curve (b) 16.9. (Accept reasonable readings.) (c) 34.4

7. 67.8

8. 11

9. (a) Same mean volume, but Y's fills vary far more (b) Machine X, with the smaller standard deviation

10. (a) $0.61x + 7.7$ (b) $\$29.1$ (c) 36.5 m^2

11. (a) 8 (4th of 7 ordered values) (b) 7.5 (c) 7.33)

12. 79

13. (a) 8 (b) 75 (c) The median

14. (a) $-2x + 31$ (b) 23

15. (a) 17 (b) 87.5 (c) B has the higher median and the smaller spread (d) Yes, possibly: the summary is symmetric

16. (a) Proof (b) $10 \leq t < 20$ (frequency 18) (c) $10 \leq t < 20$ (d) 9.42 min

17. (a) 20.4 (b) 80.4 (c) 69.2

18. (a) 0.998 (3 s.f.). A very strong positive linear correlation: hotter days are associated with more sales (b) $7.06x - 15.7$ (3 s.f.) (c) About 7 more ice creams sold per extra degree (d) 154 ice creams (e) $40\,^\circ$C lies far outside the data range

19. (a) 15 (b) 9 (c) Predicting x requires the x on y line

20. (a) Systematic (b) Stratified (c) Convenience (d) Monday-morning shoppers are not representative of all customers (e) 30

21. 15

22. (a) -0.92: strong negative; -0.35: weak negative; 0.08: no linear correlation; 0.85: strong positive (b) Correlation is not causation; a third variable is physical inactivity (c) No effect on r

23. (a) Five-number summary: $5.4,\ 8.55,\ 16.15,\ 22.6,\ 26.9$ (b) Proof (c) $\bar{x} = 15.83\,^\circ$C, agreeing with the published value (d) Roughly symmetric; it shows variation between months, not between days

24. (a) $\bar{x}$. Unchanged (b) 0 (all values already equal)

25. Sets: $\{4,4,7,8,17\}, \{4,4,7,9,16\}, \{4,4,7,10,15\}, \{4,4,7,11,14\}, \{4,4,7,12,13\}$

26. (a) 8.75 (b) 0.917

27. σ_x. The constant cancels in every deviation, leaving the spread unchanged

CHALLENGE QUESTIONS

28. (a) 7 (3rd of 5 ordered values) (b) Mean $= \frac{24+k}{5}$, median $= 7$ (c) 76 (d) The third ordered value is always 7 (e) The median is resistant; the mean is not (f) Proof

29. (a) Proof (b) 54 (c) Proof (d) 6.10 (e) $24.$)

30. (a) Proof (b) Proof (c) 1, so the standard deviation is 1. Both agree (d) Student P ($z = 1.6$ versus $z = 1.125$) (e) Standardising removes differences in mean and spread between the tests

Chapter 12 · Probability

YOUR TURN

1. (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{2}{5}$ (d) $\frac{3}{10}$ (e) $\frac{1}{4}$

2. (a) 0.65 (b) 15 (c) $\frac{3}{4}$

3. (a) $\frac{1}{2}$ (b) 0.24 (c) 60 (d) The coins are fair and symmetric; the spinner is not

4. (a) $\frac{7}{12}$ (b) 0.82 (c) $\frac{15}{16}$

5. (a) 0.7 (b) 0.9 (c) 0.1 (d) They are mutually exclusive

6. (a) $\frac{11}{26}$ (b) $\frac{7}{8}$ (c) $\frac{3}{5}$ (d) 17 (e) $\frac{7}{30}$

7. (a) $\frac{1}{6}$ (b) $\frac{5}{18}$ (c) $\frac{1}{6}$

8. (a) $\frac{3}{20}$ (b) $\frac{7}{16}$ (c) $\frac{47}{80}$

9. (a) 0.4 (b) 0.18 (c) 0.28 (d) $\frac{1}{6}$

10. (a) 0.2 (b) Not independent (c) $\frac{1}{221}$

11. (a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) The two agree, so the events are independent (d) $\frac{1}{3}$

12. (a) $\frac{9}{25}$ (b) $\frac{3}{10}$ (c) $\frac{8}{15}$

13. (a) 0.81 (b) 0.25 (c) 0.421 (d) 0.410

14. (a) $\frac{1}{6}$ (b) $\frac{3}{10}$ (c) 0.815 (d) 0.185

15. (a) 0.72 (b) 0.162 (c) 0.625

16. (a) 0.885 (b) $\frac{2}{3}$ (c) 0.862

17. (a) 0.022 (b) 0.409 (c) 0.227 (d) 0.005 is smaller than 0.009 and 0.008, so the large share is more than cancelled by the low rate

END-OF-CHAPTER PROBLEMS

1. (a) 8 (b) $\frac{6}{25}$ (c) $\frac{2}{7}$

2. (a) 0.3 (b) 0.48 (c) $0.30 \neq 0.27$, so they are **not** independent

3. (a) Tree diagram: $\frac{7}{10}, \frac{3}{10}$; then $\frac{6}{9}, \frac{3}{9}$ and $\frac{7}{9}, \frac{2}{9}$ (b) $\frac{7}{15}$ (c) $\frac{8}{15}$ (d) $\frac{8}{15}$

4. (a) 0.1875 (b) 0.24 (c) Most positive results are false alarms

5. (a) 0.65 (b) 0.8 (c) 0. Both have non-zero probability, so the two conditions contradict each other

6. (a) $\frac{1}{6}$ (b) $\frac{11}{36}$ (c) $\frac{2}{11}$

7. (a) 0.034 (b) 0.441

8. 0.5

9. (a) 0.4096 (b) 0.5904

10. 0.417

11. (a) 7 (b) $\frac{7}{40}$ (c) $\frac{4}{5}$

12. (a) 0.38 (b) 0.6 (c) 9.6

13. (a) 0.10 (b) 0.35 (c) 0.25 (d) $P(A)$.)

14. (a) A 4×6 grid of the 24 equally likely outcomes (b) $\frac{1}{8}$ (c) $\frac{1}{4}$ (d) $\frac{1}{3}$

15. (a) Tree diagram: $0.85,\ 0.15$; then $0.7,\ 0.3$ and $0.4,\ 0.6$ (b) 0.655 (c) 0.908

16. (a) $\frac{1}{5}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$

17. (a) $P(A)$: knowing B leaves $P(A)$ unchanged, so A and B are independent (b) 0.6 (c) 0.6

18. (a) Rows Boy/Girl, columns Left/Right: $10, 30, 40$; $6, 34, 40$; totals $16, 64, 80$ (b) $\frac{17}{40}$ (c) $\frac{5}{8}$ (d) 0.2. These differ, so **not** independent

19. (a) 0.059 (b) 0.508

20. (a) 0.15 (b) 0.75 (c) $\frac{6}{11}$

21. (a) 0.081 (b) 0.410 (c) 29

22. (a) $\{BB,\ BG,\ GB,\ GG\}$, equally likely (elder listed first) (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) The conditioning events differ: three outcomes against two

23. (a) 80 (b) 20 (c) 25

24. (a) 0.167 (b) 0.798 (c) It raises the probability from about 17% to 80%

25. Proof

26. $\frac{5}{14}$

27. (a) 20 (b) 0.1 (c) $\frac{4}{9}$ (d) C and T are not independent (e) 0.00955 (f) Larger: $0.01 > 0.00955$

28. (a) Tree diagram: $0.06,\ 0.94$; then $0.95,\ 0.05$ and $0.02,\ 0.98$ (b) 0.0758 (c) 0.248 (d) 0.00325 (e) 15 (f) Well for one customer, but about 15 defectives still pass

29. (a) 0.7 (b) 0.233 (c) 0.992 (d) Only 3 batteries are flat; the fourth must work (e) 1

CHALLENGE QUESTIONS

30. (a) $\frac{364}{365}$ (b) Proof (c) $\prod_{k=1}^{n-1} \frac{365-k}{365}$ (d) 50: 0.970 (3 d.p.) (e) 23, where it is 0.5073 (f) A group of 23 contains 253 pairs, not 23

31. (a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$. Switching is better: it wins twice as often (d) The $\frac{2}{3}$ transfers entirely to the one remaining door (e) Proof

32. (a) Proof (b) 0.4914 (4 d.p.) (c) Bet 1; the two probabilities differ by under 0.03 (d) 0.5086, and the two sit on opposite sides of $\frac{1}{2}$ (e) 25 rolls

Chapter 13 · Probability Distributions

YOUR TURN

1. (a) 0.3 (b) 0.40 (c) 0.1 (d) $\frac{3}{10}$ (e) 0.7

2. (a) $\frac{13}{18}$ (b) 0.3 (c) 0.6 (d) $\frac{1}{14}$

3. (a) 2.3 (b) 1.6 (c) 0.5, an expected gain of \$0.50 (d) Expected loss of \$0.25; the game is not fair

4. (a) 0.84 (b) 18

5. (a) Expected win \$1.40; a fair stake is \$1.40 (b) $-\$30$, an expected loss of \$30 (c) 14

6. (a) 0.273 (b) 0.233 (c) 0.544 (binomial cdf) (d) 0.968

7. (a) 1.8 (b) 3.6

8. (a) 0.282 (b) 0.876 (c) 2 (d) 0.205 (e) The trials are not independent and p is not constant

9. (a) $\frac{1}{9}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) 2.67 (e) 0.794

10. (a) $\frac{3}{32}$ (b) 2 (c) 2 (or by integrating $\int_0^4 x\,f(x)\,dx$) (d) 0.75 (e) 1

11. (a) 0.841 (b) 0.0912 (c) 0.683 (d) 0.933 (e) 0.0228

12. (a) 0.0548 (b) ≈ 0.683, agreeing with the 68% rule (c) 43.8, so about 44 packets (d) $\mu \pm 2\sigma$, that is from 450 g to 550 g

13. (a) 2 (b) 78.2 (c) 90.4

14. (a) 54.2 (b) 35.0 (c) 14.8

15. (a) $z_A = 1$, $z_B \approx 1.33$; B is the better performance (b) 27.5 (c) 10.8

END-OF-CHAPTER PROBLEMS

1. (a) 0.4 (b) 2.6 (c) 0.84

2. (a) \$1.10 (b) Loss of \$0.90 per ticket; the raffle favours the charity

3. (a) $X \sim B(15, 0.08)$ (b) 0.286 (c) 0.340 (d) 1.2

4. (a) 5 (b) 0.192 (c) 0.224

5. (a) $\frac{2}{9}$ (b) $\frac{4}{9}$ (c) 2 (d) 3

6. (a) 0.106 (b) 0.864 (c) 0.106, so about 10.6%

7. (a) 603 (b) 632

8. (a) 0.6745 (b) 59.7

9. (a) 9 (b) 4

10. (a) 0.106 (b) 3.17 (c) 0.964

11. (a) 1. Valid (b) $\frac{3}{4}$ (c) 0

12. (a) 0.3 (b) 0.625

13. (a) 2.60 dollars (b) $\$2.60$ (c) $-\$0.40$, so over 100 plays the expected loss is $\$40$

14. (a) 0.85; each trial has exactly two outcomes (b) 0.292 (c) 0.736 (d) 1.24

15. (a) 0.254 (b) 11

16. (a) 0.3 (b) 0.267 (c) 2.1

17. (a) 0.0912 (b) 0.656 (c) 912

18. (a) 188 g (b) 127 g (c) 0.87

19. (a) 4.82 min (b) 0.15

20. (a) $\mu + 1.645\sigma$ (b) 322 g

21. (a) Proof (b) 2; zero elsewhere (c) $\frac{1}{12}$ (d) $\frac{3}{4}$ (e) $\frac{7}{9}$

22. (a) Proof (b) $\frac{1}{2}$, a maximum (c) $\frac{1}{2}$ (d) $\frac{1}{20}$

23. (a) 0.841 (b) 0.401 (c) 0.756

24. (a) Proof (b) 0.263 (c) 0.254, close to the modelled 0.263; the normal model is a little high here (d) 0.668, or 66.8%, against the 68.3% the normal model predicts (e) A good fit, to within about one percentage point

CHALLENGE QUESTIONS

25. (a) W takes $20, 5, 0$ with probabilities $\frac{1}{36}, \frac{5}{36}, \frac{30}{36}$ (b) $\$1.25$ (c) $\$1.25$; fair means the expected gain is zero (d) $\$375$ (e) 3.61 (f) Chance dominates ten games; the stall's average approaches $\$0.75$

26. (a) 3.54 (b) 0.8811 (binomial cdf) (c) 30 (d) ≈ 0.8802, matching 0.8811 to three decimal places (e) 0.6572, close to 0.6563 (f) Use the continuity correction; accuracy improves as n grows

27. (a) Proof (b) Expanding: $np - rp + p \geq r - rp$, so $np + p \geq r$, that is $r \leq (n+1)p$ (c) 6.3, so the mode is 6 (d) 0.246, and both 4 and 5 are modes

Chapter 14 · Limits & Derivatives

YOUR TURN

1. (a) 5 (b) 5 (c) 0 (d) $\frac{1}{2}$

2. (a) $x + 4 \to 8$ (b) $x + 1 \to 2$ (c) $x + 1 \to 3$ (d) $x - 3 \to -6$

3. 1

4. The one-sided limits disagree, so there is no single limit

5. (a) Converges to 0 (b) Diverges: oscillates between -1 and 1 without approaching a value (c) Diverges: grows without bound (d) $3 + \frac{1}{x} \to 3$: converges to 3

6. The limit appears to be $\frac{1}{2}$

7. (a) $2x + h \to 2x$ (b) $6x + 3h \to 6x$ (c) $2x + h + 1 \to 2x + 1$ (d) $2x + h - 3 \to 2x - 3$

8. 0

9. e^x

10. $f'(x)$, $\frac{dy}{dx}$, $\frac{d}{dx}f(x)$

11. The graph has a corner: unbroken, but not smooth

12. (a) $7x^6$ (b) $15x^2 - 2$ (c) $4x^3 - 6x$ (d) $20x^4 + 1$ (e) $6x^2 - 14x + 1$ (f) 0

13. (a) $2x^{-1} \Rightarrow -\frac{2}{x^2}$ (b) $x^{1/2} \Rightarrow \frac{1}{2\sqrt{x}}$ (c) $x^3 + x^{-1} \Rightarrow 3x^2 - \frac{1}{x^2}$ (d) $x^{-3} \Rightarrow -\frac{3}{x^4}$ (e) $x^{1/3} \Rightarrow \frac{1}{3\sqrt[3]{x^2}}$ (f) $x + x^{-1} \Rightarrow 1 - \frac{1}{x^2}$

14. (a) $2x - 5$, so 1 (b) $3x^2 - 4$, so -1 (c) $\frac{1}{2\sqrt{x}}$, so $\frac{1}{4}$ (d) $6x^2 + 1$, so 7

15. (a) 3 (b) ± 2

16. $x^{1/4}$, giving $\frac{1}{4}x^{-3/4}$

17. (a) $4x - 4$ (b) $-2x + 2$ (c) $-x + 2$ (d) $x + 1$

18. (a) $-\frac{1}{3}(x - 1)$ (b) $-4(x - 4)$

19. a; its gradient is undefined

20. ± 2

21. -9, so $(3, -9)$

22. (a) $\cos x - \sin x$ (b) $e^x - \frac{3}{x}$ (c) $-4\sin x$ (d) $\sec^2 x$ (e) $\frac{1}{1+x^2}$ (f) $3^x \ln 3$

23. (a) $2\cos x - 5e^x$ (b) $\frac{1}{x} + \sec^2 x$ (c) $\sec x \tan x$ (d) $\frac{1}{\sqrt{1-x^2}}$

24. $\frac{1}{2}$

25. $x + 1$

26. In degrees the limit is $\frac{\pi}{180}$, not 1

27. (a) $8(2x+1)^3$ (b) $2x\cos(x^2)$ (c) $3e^{3x}$ (d) $\frac{3}{2\sqrt{3x+1}}$

28. (a) $\cos x - x\sin x$ (b) $xe^x(x+2)$ (c) $x^2(3\ln x + 1)$ (d) $\sin x + x\cos x$

29. (a) $\frac{1}{(x+1)^2}$ (b) $\frac{-2}{(x-1)^2}$ (c) $\frac{x\cos x - \sin x}{x^2}$ (d) $\frac{e^x(x-1)}{x^2}$

30. $u = v = x$: $(uv)' = 2x$ but $u'v' = 1$

31. 108

32. Correct: $3\cos(3x)$

33. (a) $6x - 12$ (b) $4x^3$, then $12x^2$ (c) $\cos x$, then $-\sin x$ (d) $2e^{2x}$, then $4e^{2x}$

34. (a) $2x - 4 > 0 \Rightarrow x > 2$ (b) Testing the four intervals: $-1 < x < 0$ or $x > 1$

35. (a) $6x > 0 \Rightarrow x > 0$ (b) $6x - 6 > 0 \Rightarrow x > 1$

36. $(1, -2)$; f'' changes sign at $x = 1$

37. 3: minimum

38. $\frac{1}{2}$ second

39. (a) $\frac{0}{0}$; $\frac{e^x}{1} \to 1$ (b) $\frac{0}{0}$; $\frac{2\cos 2x}{1} \to 2$ (c) $\frac{0}{0}$; $\frac{\sec^2 x}{1} \to 1$ (d) $\frac{1}{2x^2} \to 0$

40. (a) $\frac{0}{0} \to \frac{\sin x}{2x}$, still $\frac{0}{0} \to \frac{\cos x}{2} \to \frac{1}{2}$ (b) $\frac{0}{0} \to \frac{e^x - 1}{2x}$, still $\frac{0}{0} \to \frac{e^x}{2} \to \frac{1}{2}$

41. Both give $\frac{1}{2}$

42. The correct value is 2

43. $\frac{0}{0}$ and $\frac{\infty}{\infty}$; $\frac{2}{0}$ is not indeterminate

END-OF-CHAPTER PROBLEMS

1. 1

2. $6x - 9$

3. 2

4. $3x^2$

5. Proof

6. (a) $3x^2 - 3$ (b) ± 1. Points: $(1, -2)$ and $(-1, 2)$

7. (a) 1.0517 (b) 1 (c) 1

8. $6x - 2$

9. (a) V-shaped graph with vertex at $(2, 0)$ (b) $f(2)$: no hole or jump (c) $\frac{|h|}{h}$ is 1 for $h > 0$ and -1 for $h < 0$; the one-sided limits disagree, so no derivative exists (a corner)

10. $\frac{9}{2}$

11. Points: $(2, 2)$ and $(-2, -2)$

12. (a) $\frac{2e^{2x}(x^2 - x + 1)}{(1+x^2)^2}$ (b) 2

13. $\frac{1}{6}$

14. (a) $2x - 2$ (b) 4

15. (a) 3, so the point is $(3, 3)$ (b) $4x - 9$ (c) $-\frac{1}{4}(x - 3)$

16. (a) $\frac{1}{2}$ (b) $\frac{1}{2}$

17. (a) $\frac{1 - \ln x}{x^2}$ (b) 0, so the tangent is horizontal

18. (a) $24x^2(2x^3 - 1)^3$ (b) $x(2\ln x + 1)$ (c) $\frac{x\cos x - \sin x}{x^2}$

19. (a) Proof (b) $-\frac{1}{e}(x - 1)$

20. (a) $\frac{3}{1+9x^2}$ (b) $2\sec(2x)\tan(2x)$ (c) $\frac{1}{x\ln 2}$

21. (a) $3(x-1)(x-3) > 0 \Rightarrow x < 1$ or $x > 3$ (b) $6x - 12 > 0 \Rightarrow x > 2$ (c) 2

22. $-2x - 1$

23. (a) $30x(3x^2+1)^4$ (b) $e^x(x^2 + 2x)$ (c) $\frac{1}{(x+1)^2}$

24. (a) $\sec x \tan x$ (b) $-\tan x$ (c) $\arctan x + \frac{x}{1+x^2}$

25. (a) $3t^2 - 12t + 9$ (b) 3 seconds (c) 6 m s^{-2}

26. (a) $f'(x) > 0 \Rightarrow x < -2$ or $x > 4$ (b) 4 (minimum) (c) 1

27. the limit is $\frac{1}{3}$

28. (a) 3 (b) $f'(x) > 0$ for $x < -1$ or $x > 3$: increasing on these intervals (c) 1 (point of inflexion)

29. (a) $3(x-3)(x+1)$ (b) -22, so $(-1, 10)$ and $(3, -22)$ (c) $12 > 0$: minimum (d) -6, so $(1, -6)$ (e) Concave up

30. (a) 2 (ratio of leading coefficients) (b) Proof (c) $(x+2)^2 > 0$ (d) $\frac{5}{4}x - \frac{1}{2}$ (e) $-\frac{4}{5}x - \frac{1}{2}$

31. (a) Proof (b) $(2, 0.541)$ (c) Proof (d) $-2e^{-2} < 0$, maximum (e) 0

32. (a) Proof (b) Proof (c) $4\pi r - \frac{1000}{r^2}$ (d) 4.30 cm (e) $4\pi + \frac{2000}{r^3} > 0$ for $r > 0$, so the stationary value is a minimum (f) Proof

CHALLENGE QUESTIONS

33. (a) $x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots$ (b) Proof (c) Proof (d) $4x^3h + 6x^2h^2 + 4xh^3 + h^4$; dividing by h gives $4x^3 + 6x^2h + 4xh^2 + h^3 \to 4x^3$ (e) $-x^{-2}$, which agrees

34. (a) 0.99833, 0.99998, 1.00000, and the same for negative x (b) Proof (c) Proof (d) Proof (e) The clean formula holds only in radians

35. (a) Three applications, each checked indeterminate first (b) Dividing by x^3 gives $\frac{1}{6} - \frac{x^2}{120} + \cdots \to \frac{1}{6}$ (c) $\frac{x^2}{2} + \frac{x^3}{6} + \cdots$, so the fraction is $\frac{1}{2} + \frac{x}{6} + \cdots \to \frac{1}{2}$ (d) L'Hôpital: general. Series: one step. Harder: $\lim_{x\to\infty} \frac{\ln x}{x}$

Chapter 15 · Derivative Applications

YOUR TURN

1. (a) -4, so $(3,-4)$ (b) ± 2, giving $(2,-16)$ and $(-2,16)$ (c) Points $(2,-19)$ and $(-1,8)$ (d) Points $(0,0)$, $(1,-1)$, $(-1,-1)$
2. (a) $2>0$, so $(3,-4)$ is a local minimum (b) -2 it is $-12<0$, a maximum (c) $2>0$, a local minimum at $(1,2)$ (d) 4: minimum
3. (a) 0 is stationary (b) The test gives no conclusion when the second derivative is zero (c) Stationary point of inflexion
4. (a) -11, so $(2,-11)$ (b) 2, so $(-1,2)$ (c) $\frac{9}{2}$, so $(\frac{3}{2},\frac{9}{2})$
5. (a) 0 is a minimum (b) This is a genuine point of inflexion (c) $6x-12<0$ when $x<2$, so the curve is concave down on $x<2$
6. (a) 100 (b) The rectangle is a 10×10 square (c) 12
7. (a) Need $x>0$ and $12-2x>0$, so $0<x<6$ (b) 2 gives the maximum (c) 128 cm^3
8. (a) 1250 m^2 (b) $2\pi+\frac{4000}{r^3}>0$, a minimum (c) 5
9. (a) 2 m s^{-2} (b) 2 s (c) 6 m s^{-2} (d) 1 m s^{-2}
10. (a) 3 s (b) 6 m s^{-2} (c) The signs are the same, so the particle is speeding up (d) The signs differ, so the particle is slowing down
11. (a) $-\frac{x}{y}$ (b) $-\frac{y}{x}$ (c) $\frac{x}{y}$ (d) $-\frac{x^2}{y^2}$
12. (a) $-\frac{2x+y}{x}$ (b) $-\frac{2xy}{x^2+2y}$
13. (a) $\frac{4}{3}$ (b) $4-x$ (c) 25, giving $(0,5)$ and $(0,-5)$ (d) Naming y as well selects which point on the curve is meant
14. (a) 20 cm^{2}s^{-1} (b) 75.4 cm^{2}s^{-1} (c) 300 cm^{3}s^{-1} (d) 2.51 cm s^{-1}
15. (a) 0.0398 cm s^{-1} (b) $\frac{dy}{dt}=-1.5$ m s^{-1} (c) 0.127 cm s^{-1}

END-OF-CHAPTER PROBLEMS

1. Proof
2. 25.1 m^{2}s^{-1}
3. (a) $1, 3$ s (b) -3 m s^{-1} (minimum velocity)
4. 800 dollars
5. -2
6. (a) $-\frac{2x+y}{x+2y}$ (b) $-x+2$
7. (a) $\frac{1}{3}\pi h^3$ (b) $\pi h^2\frac{dh}{dt}$
8. (a) -2 (b) 1
9. (a) 4 (b) 1
10. (a) 15 (b) $\frac{2-x}{y+1}$ (c) $3+\frac{1}{2}x$
11. 0.675 m s^{-1}
12. 0.212 m min^{-1}
13. 1.94
14. Proof
15. (a) $2\pi r^2+\frac{1000}{r}$ (b) $4\pi+\frac{2000}{r^3}>0$: minimum
16. (a) $2x+\frac{200}{x}$ (b) 20. ($L''>0$: minimum.) Dimensions 10×20 m
17. (a) -6 (b) $-9<0$ maximum
18. (a) $x^2+\frac{128}{x}$ (b) 4 m. ($A''>0$: minimum.)
19. (a) Proof (b) 4000 cm^3
20. Proof
21. (a) -0.375 m s^{-1} (b) 0.4375 m^{2}s^{-1}
22. 4.24 units s^{-1}
23. (a) 3: min $(3,-22)$ (b) -6. Inflexion $(1,-6)$ (c) $f'<0$ between the roots: decreasing on $-1<x<3$
24. (a) 3 s (b) -6 m s^{-2} (c) $-3<0$. Same sign, so the particle is speeding up
25. (a) $6>0$, min at $(3,2)$ (b) $-3\neq 0$: a non-stationary inflexion
26. (a) $2, 4$ s (b) 6 m s^{-2} (c) $2<t<3$ and $4<t\leq 6$
27. (a) 1: point $(1,\frac{1}{e})$ (b) $-e^{-1}<0$: a local maximum (c) 2
28. (a) $4x^2(x-3)$ (b) -27, so $(0,0)$ and $(3,-27)$ (c) $36>0$: minimum (d) It is neither a maximum nor a minimum: a stationary inflexion (e) -16, so $(0,0)$ and $(2,-16)$
29. (a) $0<x<6$ (b) 2 (c) 6 is excluded by the domain (d) 128 cm^3
30. (a) $6t-12$ (b) 3 seconds (c) 0 m (d) -3 m s^{-1} (e) Same signs, so the particle is speeding up
31. (a) 72: the point lies on C (b) Proof (c) $\frac{4}{5}$ (d) 0

CHALLENGE QUESTIONS

32. (a) Proof (b) Proof (c) $4\pi + \frac{4V}{r^3} > 0$ (d) Proof (e) Waste in cutting the ends; the seam adds material

33. (a) Proof (b) Using D^2 avoids differentiating a square root (c) 1.94 (d) Proof (e) The shortest segment meets the curve along the normal

34. (a) Proof (b) 2 (c) Far from the vertex a parabola is almost straight (d) Proof (e) $\rho = \frac{1}{2}$; the circle $x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4}$

Chapter 16 · Integral Calculus

YOUR TURN

1. (a) $x^4 - 3x^2 + C$ (b) $\frac{x^3}{3} + \frac{3x^2}{2} - 5x + C$ (c) $-\frac{3}{x^2} + C$ (d) $\frac{2}{3}x^{3/2} + C$ (e) $\frac{x^3}{3} + \frac{1}{x} + C$ (f) $\frac{x^3}{3} - 2x + C$ (g) $2\sqrt{x} + C$ (h) $2x^4 + C$

2. (a) $2x^2 - 3x + 3$ (b) $2x^{3/2} + 4$ (c) $x^3 + x + 2$ (d) A point on the curve is needed to fix C (e) Two integrations introduce two arbitrary constants

3. (a) 6 (b) 12 (c) 1 (d) 2 (e) 1 (f) 6

4. (a) $\frac{26}{3}$ (b) $\frac{4}{3}$ (c) 2 (d) 4 (e) 8

5. (a) 1.75 (b) 3.75 (c) $\frac{8}{3} \approx 2.67$ (d) 2.1875, closer to $\frac{8}{3}$ than the four-rectangle value 1.75 (e) Both converge to $\frac{8}{3}$

6. (a) $2\sin x + \cos x + C$ (b) $3e^x + \ln|x| + C$ (c) $\frac{1}{3}e^{3x} + C$ (d) $\frac{(3x-2)^5}{15} + C$ (e) $-\frac{1}{2}\cos 2x + C$ (f) $\frac{1}{4}\tan 4x + C$ (g) $\frac{1}{3}\sin(3x + 1) + C$ (h) $\frac{1}{2}\ln|2x + 5| + C$

7. (a) 3, so $\frac{1}{3}\arctan\frac{x}{3} + C$ (b) 4, so $\arcsin\frac{x}{4} + C$ (c) $\frac{5^x}{\ln 5} + C$ (d) $\frac{1}{3}\arctan\frac{x+2}{3} + C$ (e) $\frac{1}{2}\arcsin\frac{2x}{3} + C$ (f) $\arctan\frac{x}{2} + C$

8. (a) $\frac{1}{6}\ln\left|\frac{x-3}{x+3}\right| + C$ (b) $\ln|x + 1| + 3\ln|x + 2| + C$ (c) Factorises: partial fractions. No real roots: complete the square

9. (a) $x + 3$, so the integral is $\frac{x^2}{2} + 3x + C$ (b) $2 + \frac{1}{x}$, so the integral is $2x + \ln|x| + C$ (c) $\frac{x^2}{2} + \frac{1}{x} + C$ (d) $\frac{1}{2}\arctan\frac{x+3}{2} + C$

10. (a) $\frac{(x^3+1)^5}{5} + C$ (b) $\frac{1}{2}e^{x^2} + C$ (c) $\frac{2}{3}(x^2 + 1)^{3/2} + C$ (d) $-\frac{1}{2(x^2+4)} + C$ (e) $\frac{1}{2}(\ln x)^2 + C$ (f) $\frac{1}{4}\sin^4 x + C$

11. (a) $\frac{31}{5}$ (b) $\frac{31}{5}$, the same (c) $\frac{1}{3}$

12. (a) $\cos x$: $x\sin x + \cos x + C$ (b) e^{2x}: $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$ (c) x: $\frac{x^2}{2}\ln x - \frac{x^2}{4} + C$ (d) $x\arctan x - \frac{1}{2}\ln(1 + x^2) + C$ (e) $\frac{x^3}{3}\ln x - \frac{x^3}{9} + C$ (f) $x\tan x + \ln|\cos x| + C$

13. (a) π (b) 1 (c) e^x would leave $\int \frac{x^2}{2}e^x\,dx$, with a higher power than before (d) $\sin x$ (e) Neither factor simplifies; the integral returns to itself

14. (a) $\frac{1}{6}$ (b) $\frac{32}{3}$ (c) $\frac{32}{3}$ (d) 9 (e) $\frac{8}{3}$

15. (a) 2 (b) $x + 2$ is above the parabola between those values (c) $\frac{9}{2}$ (d) $-\frac{9}{2}$; the curves were subtracted the wrong way round

16. (a) $\frac{32\pi}{5}$ (b) 8π (c) 36π (d) $\frac{2\pi}{3}$ (e) $\frac{\pi^2}{2}$

17. (a) 8π (b) The two rotations sweep out different solids (c) $\frac{32\pi}{5}$

18. (a) 12 m (b) $3t^2 + 1$ (c) -3 m (d) 8 m (e) 0 m; the particle returns to its starting point (f) 23 m

END-OF-CHAPTER PROBLEMS

1. (a) $x^3 - 2x^2 + x + 2$ (b) 4

2. (a) ± 2 (b) $\frac{32}{3}$

3. (a) 4. Points $(\pm 2, 4)$ (b) $\frac{64}{3}$

4. (a) $\frac{1}{2}\ln(x^2 + 4) + C$ (b) $\frac{1}{2}\ln 2$

5. (a) $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$ (as in Q29) (b) $\frac{1}{4}e^2 + \frac{1}{4}$

6. (a) $\displaystyle\int_0^4 \pi x\,dx$ (b) 8π

7. (a) $3t^2 - 4t + 5$ (b) $t^3 - 2t^2 + 5t$ (c) 10 m

8. (a) $\frac{16}{3}$ (b) 8π

9. (a) $\frac{1}{2}\arctan\frac{x}{2} + C$ (b) $\frac{\pi}{8}$

10. (a) $1, 3$ s (b) 8 m

11. (a) $2x^2 - 3x + 3$ (b) y-intercept 3

12. (a) $\frac{1}{3}e^{3x} + C$ (b) $\frac{(4x-1)^7}{28} + C$ (c) $-\frac{1}{2}\cos 2x + C$

13. 7

14. (a) 1 (b) 2

15. $\frac{9}{2}$

16. 78

17. $\frac{1}{2}\ln 2$

18. $2\ln|x - 1| - \ln|x + 3| + C$

19. $\frac{1}{3}\arctan\left(\frac{x-2}{3}\right) + C$

20. (a) $x\sin x + \cos x + C$ (b) $\frac{x^3}{3}\ln x - \frac{x^3}{9} + C$

21. (a) 8π (b) $\frac{32\pi}{5}$

22. (a) $\frac{4}{3}$ m (b) 4 m

23. (a) 8 (b) $1, 2, 3$ are $2, 5, 10$, so the estimate is 17 (c) $8 < 12 < 17$ (d) Increase, towards 12

24. (a) 1 (b) Region between them shaded (c) $\frac{9}{2}$ (d) The line is the upper boundary throughout

25. (a) $\frac{x^2}{2}\ln x-\frac{x^2}{4}+C$ (b) $\frac{e^2+1}{4}$ (c) $\frac{\pi}{2}-1$ (d) $\cos x$ would leave $\int \frac{x^2}{2}\sin x\,dx$, with a higher power of x than before

26. (a) $\frac{\pi}{7}$ (b) $\frac{3\pi}{5}$ (c) The second, $\frac{3\pi}{5}$; its region lies further from its axis

27. (a) $6t-t^2$ (b) 6 s (c) 9 m s^{-1} (d) 36 m (e) The velocity never changes sign on this interval

28. (a) $\frac{x^2}{2} - x + 2\ln|x+1| + C$ (b) $\ln 3$ (c) $\frac{1}{3}$ (d) $\frac{1}{3}\arctan\frac{x-2}{3}+C$

CHALLENGE QUESTIONS

29. (a) A cylinder of radius r and height h (b) $\frac{1}{3}\pi r^2 h$ (c) $\frac{4}{3}\pi r^3$ (d) From integrating x^2 to give $\frac{x^3}{3}$

30. (a) $\frac{1}{2}e^x(\sin x+\cos x)+C$ (b) I, which is true but gives no information (c) $e^x(x^2-2x+2)+C$; the power drops each round (d) Four rounds

31. (a) 1 (b) $\frac{n-1}{n}I_{n-2}$ (c) $\frac{8}{15}$ (d) Even n traces back to $I_0=\frac{\pi}{2}$, odd to $I_1=1$

Chapter 17 · Differential Equations

YOUR TURN

1. (a) x^2+C (b) Ae^x (c) $A(1+x)-1$ (d) $\ln(e^x+C)$

2. (a) $2e^{x^2/2}$ (b) e^{x^3} (c) $4\sec x$ (d) One integration gives one constant; the initial condition fixes it

3. (a) $x(2\ln|x|+C)$ (b) $x(C-\ln|x|)$ (c) $x(Ax-1)$ (d) Cx

4. (a) Proof (b) $x^2(2\ln|x|+C)$

5. (a) $4+Ce^{-x}$ (b) $3+Ce^{-2x}$ (c) $\frac{1}{3}e^x+Ce^{-2x}$ (d) $e^{2x}+Ce^x$

6. (a) x^3 (b) $\frac{x^2}{4}+\frac{C}{x^2}$ (c) $\frac{x^2}{3}+\frac{2}{3x}$ (d) It makes the left side $\frac{d}{dx}(Iy)$

7. (a) 1.5 (b) 1.21 (c) 1.48 (d) 1.221

8. (a) 2.4 (b) 0.25 (c) Euler gave 1.21, which is about 0.01 too small (d) e^x is concave up, so each straight step lies below the curve and the shortfall accumulates

9. (a) $1+2x+2x^2$ (b) $1-\frac{x^2}{2}+\frac{x^4}{24}$ (c) $2x-2x^2+\frac{8x^3}{3}$ (d) $1-x^2+\frac{x^4}{2}$

10. (a) $2x-\frac{4x^3}{3}$ (b) Substitute x^2 into the sine series: $x^2-\frac{x^6}{6}$ (c) Substitute $2x$ into the arctangent series: $2x-\frac{8x^3}{3}$ (d) Binomial

11. (a) 1.105 (b) 0.9801 (4 d.p.) (c) The cosine estimate is far more accurate (d) $1+x-\frac{x^3}{3}$

12. (a) 1 (b) 2 (c) $1+x+x^2+\frac{x^3}{3}+\cdots$ (d) 1.399; closer than Euler's 1.362 to the exact 1.3997

END-OF-CHAPTER PROBLEMS

1. (a) $500e^{0.2t}$ (b) 3695

2. (a) $25+65e^{-0.1t}$ (b) 48.9°C (c) As $t\to\infty$, $\theta\to 25$°C

3. (a) $\frac{1}{v}+v$ (b) $x^2(2\ln|x|+A)$

4. (a) x (b) $x^2+\frac{1}{x}$

5. (a) 1.625 (b) 1.797 (c) Euler underestimates by about 0.17

6. (a) 1 (b) $1-\frac{x^2}{2}+\frac{x^4}{24}-\cdots$ (c) 0.9801

7. (a) $x-\frac{x^3}{6}+\frac{x^5}{120}-\cdots$ (b) $\frac{1}{6}-\frac{x^2}{120}+\cdots\to\frac{1}{6}$

8. (a) $1+x+\frac{x^2}{2}+\frac{x^3}{6}+\cdots$ (b) $1-x^2+\frac{x^4}{2}-\cdots$ (c) 0.767

9. (a) $25+65e^{-kt}$ (b) 0.0486 (c) 30.2 minutes

10. (a) Proof (b) $\frac{x}{1-\ln x}$

11. (a) $\frac{x^3}{2}+\frac{x}{2}$ (b) $e^{-x}-e^{-2x}$

12. $\sin x+2\cos x$

13. (a) Proof (b) 2.75 (time units)

14. (a) 1.55 (3 sf) (b) Below the true value

15. (a) 6.82 m s^{-1} (b) 7 m s^{-1}. Euler overshoots 7 then corrects back:

16. (a) $2x-2x^2+\frac{8x^3}{3}$ (b) $2x-2x^2+\frac{8x^3}{3}$

17. (a) $x-x^2+\frac{x^3}{3}+\cdots$ (b) 0.0903330 — the two agree to six decimal places

18. (a) $1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots$ (b) 2.90 (c) Very slow: four terms give barely one correct digit

19. Divide by x^3: $\frac{1}{6}-\frac{x^2}{120}+\cdots\to\frac{1}{6}$

20. $1+x+\frac{3x^2}{2}+\frac{4x^3}{3}+\cdots$

21. (a) e^{kt} (b) Proof (c) 14.0 mg L^{-1} (d) 20 mg L^{-1}; the concentration plateaus rather than accumulating (e) 7.68 hours (f) 2.31 hours

22. (a) $x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots$ (b) $-x-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}-\cdots$ (c) Proof (d) 1.0986.) (e) $\frac{1}{2}$, comfortably inside the valid range, and its terms shrink quickly

23. (a) Proof (b) Proof (c) 12 minutes (d) 0.36 m (e) 0.149 m, so the estimate is 41% too low (f) A decreasing, concave-up parabola; tangent steps fall below it

CHALLENGE QUESTIONS

24. (a) $20+60e^{-kt}$ (b) 0.0405 (c) 46.7° (d) Equivalently $60e^{-kt} > 0$ (e) 10 minutes, independent of the starting temperature

25. (a) $(1+h)^n$ (b) $\left(1+\frac{1}{n}\right)^n$ (c) 2.71828) (d) The error halves; Euler is a first-order method (e) e; Euler's estimate converges to the exact value

26. (a) $-k\sqrt{1+v^2}$ (b) x^{-k} (c) $\frac{1}{2}\left(x^{1-k}-x^{1+k}\right)$ (d) $(0,0)$, the dock itself (e) $k=1$: lands $\frac{1}{2}$ upstream; $k>1$: swept away

Chapter 18 · Complex Numbers

YOUR TURN

1. (a) $4-2i$ (b) $3-4i$ (c) $5+i$ (d) 5

2. (a) $-i$ (b) 1 (c) $-i$ (d) 1

3. (a) -4 (b) $7+4i$ (c) 65

4. (a) $1+2i$ (b) $\frac{1}{2}-\frac{1}{2}i$

5. (a) $2+2i$ (b) $3+2i$

6. (a) 5 (b) 13 (c) $\sqrt{2}$ (d) 2

7. (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) π (d) Third quadrant, so $-\frac{3\pi}{4}$

8. (a) $2e^{i\pi/2}$ (b) $3e^{i\pi}$ (c) $\sqrt{2}\,e^{-i\pi/4}$

9. (a) $2+2\sqrt{3}\,i$ (b) $\sqrt{2}-\sqrt{2}\,i$

10. (a) Moduli multiply, arguments add: $6\,\text{cis}\,\frac{\pi}{2}$ (b) Moduli divide, arguments subtract: $3\,\text{cis}\,\frac{\pi}{3}$ (c) -1) (d) $8i$)

11. (a) 12 (b) $\frac{5\pi}{12}$ (c) $\frac{3}{4}$ (d) $\frac{\pi}{12}$

12. (a) $\cos 4\theta + i\sin 4\theta$ (b) $-8i$

13. (a) $2\cos\theta$ (b) $2\cos 3\theta$ (c) $2i\sin 2\theta$

14. (a) $3i$ (or $-3i$) (b) 3 (c) Five (d) A regular n-gon centred at the origin

15. (a) $1,\ -\frac{1}{2}+\frac{\sqrt{3}}{2}i,\ -\frac{1}{2}-\frac{\sqrt{3}}{2}i$ (b) $1,\ i,\ -1,\ -i$

16. (a) 3: roots $\pm(3+2i)$ (b) 1: roots $\pm(1+3i)$

17. (a) $8\,\text{cis}\,0$; roots $2\,\text{cis}\,\frac{2\pi k}{3}$: $2\,\text{cis}\,0,\ 2\,\text{cis}\,\frac{2\pi}{3},\ 2\,\text{cis}(-\frac{2\pi}{3})$ (b) $16\,\text{cis}\,0$; roots $2\,\text{cis}\,\frac{\pi k}{2}$: $2\,\text{cis}\,0,\ 2\,\text{cis}\,\frac{\pi}{2},\ 2\,\text{cis}\,\pi,\ 2\,\text{cis}(-\frac{\pi}{2})$

18. (a) $\pm 3i$ (b) $\pm 2i$ (c) $1\pm 2i$ (d) $-1\pm i$

19. (a) $3-i$ (b) z^2+4 (c) z^2-2z+2

20. (a) c (b) Non-real roots come in pairs; three roots cannot pair up

END-OF-CHAPTER PROBLEMS

1. (a) $4+2i$ (b) $11-2i$ (c) $-1+2i$ (d) 5

2. $\frac{1}{2}+\frac{1}{2}i$

3. (a) $\frac{3\pi}{4}$ (b) $-\frac{\pi}{6}$

4. $2e^{i\pi/3}$

5. (a) 64 (b) $32i$ (c) -1

6. (a) 2: roots $\pm(2+i)$ (b) 2: roots $\pm(2+3i)$

7. (a) $4\,\text{cis}\,\frac{2\pi}{3}$ (b) -8

8. 5

9. (a) $2\,\text{cis}\,\frac{\pi}{6},\ 2\,\text{cis}\,\frac{5\pi}{6},\ 2\,\text{cis}(-\frac{\pi}{2})$ (b) $\sqrt{3}+i,\ \ -\sqrt{3}+i,\ \ -2i$

10. (a) -1 (b) i

11. $0,1,2$: $3\,\text{cis}\,0,\ 3\,\text{cis}\,\frac{2\pi}{3},\ 3\,\text{cis}(-\frac{2\pi}{3})$

12. 1

13. (a) $8e^{i\pi}$ (b) $\sqrt{3}+i$

14. (a) $\sqrt{2}+\sqrt{2}i,\ -\sqrt{2}+\sqrt{2}i,\ -\sqrt{2}-\sqrt{2}i,\ \sqrt{2}-\sqrt{2}i$ (b) Proof

15. $-\frac{1}{2}\pm\frac{3}{2}i$

16. Proof

17. 2

18. $z^2-4z+13$

19. $\pm(1+i)$

20. $2\sqrt{2}\,e^{i\pi/4}$

21. $3+4i$

22. Proof

23. Proof

24. Proof

25. $1-\sqrt{3}\,i$

26. (a) 1, $\text{cis}\,\frac{2\pi}{5}$, $\text{cis}\,\frac{4\pi}{5}$, $\text{cis}\,\frac{6\pi}{5}$, $\text{cis}\,\frac{8\pi}{5}$ (b) Proof (c) Proof (d) Proof (e

27. (a) Proof (b) Proof (c) Proof (d) $\frac{3\pi}{16}$ (e) It turned a fourth power into easily integrated multiple angles

28. (a) The coefficients are real, so non-real roots occur in conjugate pairs (b) z^2-2z+5 (c) The other factor is z^2+9 (d) All four roots: $1+2i,\ 1-2i,\ 3i,\ -3i$ (e) Symmetric about the real axis; $P(z)$ has real coefficients

CHALLENGE QUESTIONS

29. (a) Proof (b) Proof (c) Proof (d) -0.5000 (e) $-\frac{1}{2}$

30. (a) 1, so the modulus is unchanged and the argument increases by α — a rotation about the origin (b) Proof (c) $\frac{1}{2} - \frac{\sqrt{3}}{2}i$ (d) Proof (e) Proof

31. (a) cis $\frac{\theta}{3}$, cis $\frac{\theta+2\pi}{3}$, cis $\frac{\theta+4\pi}{3}$ (b) The left side is multi-valued; the right names one value (c) i; the roots are $\pm i$, and the formula misses $-i$ (d) Proof